

Multimedia

Vorlesung

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Based on material by Prof. Christian Breiteneder

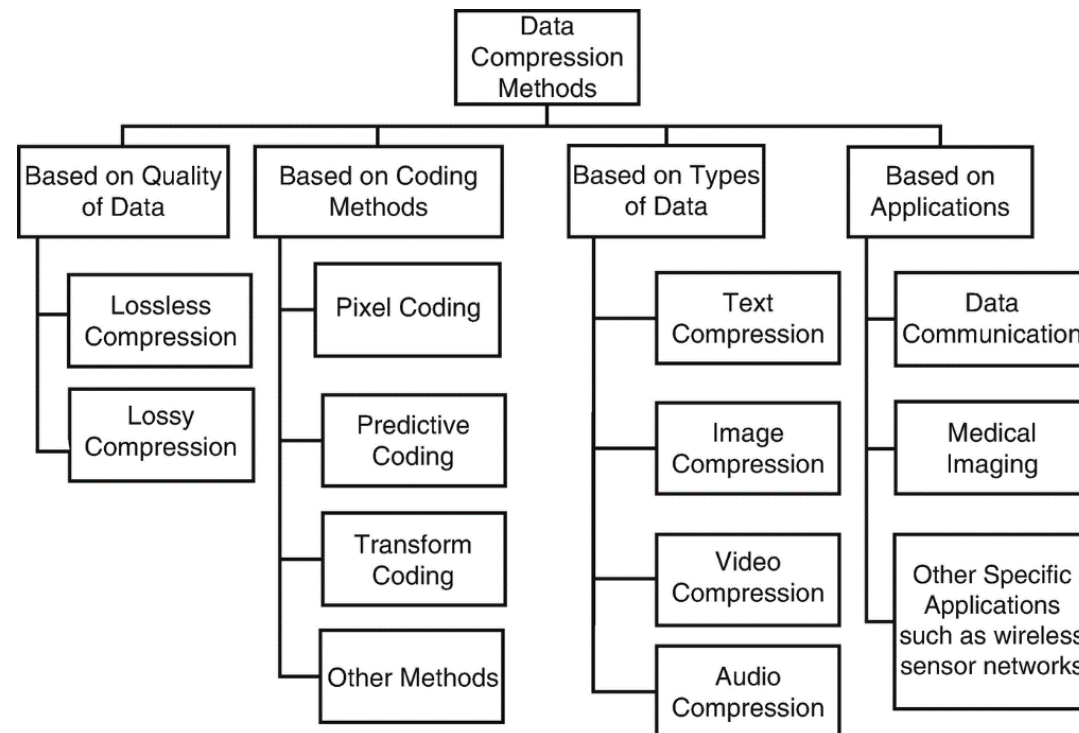
Multimedia Compression Basics

Codec

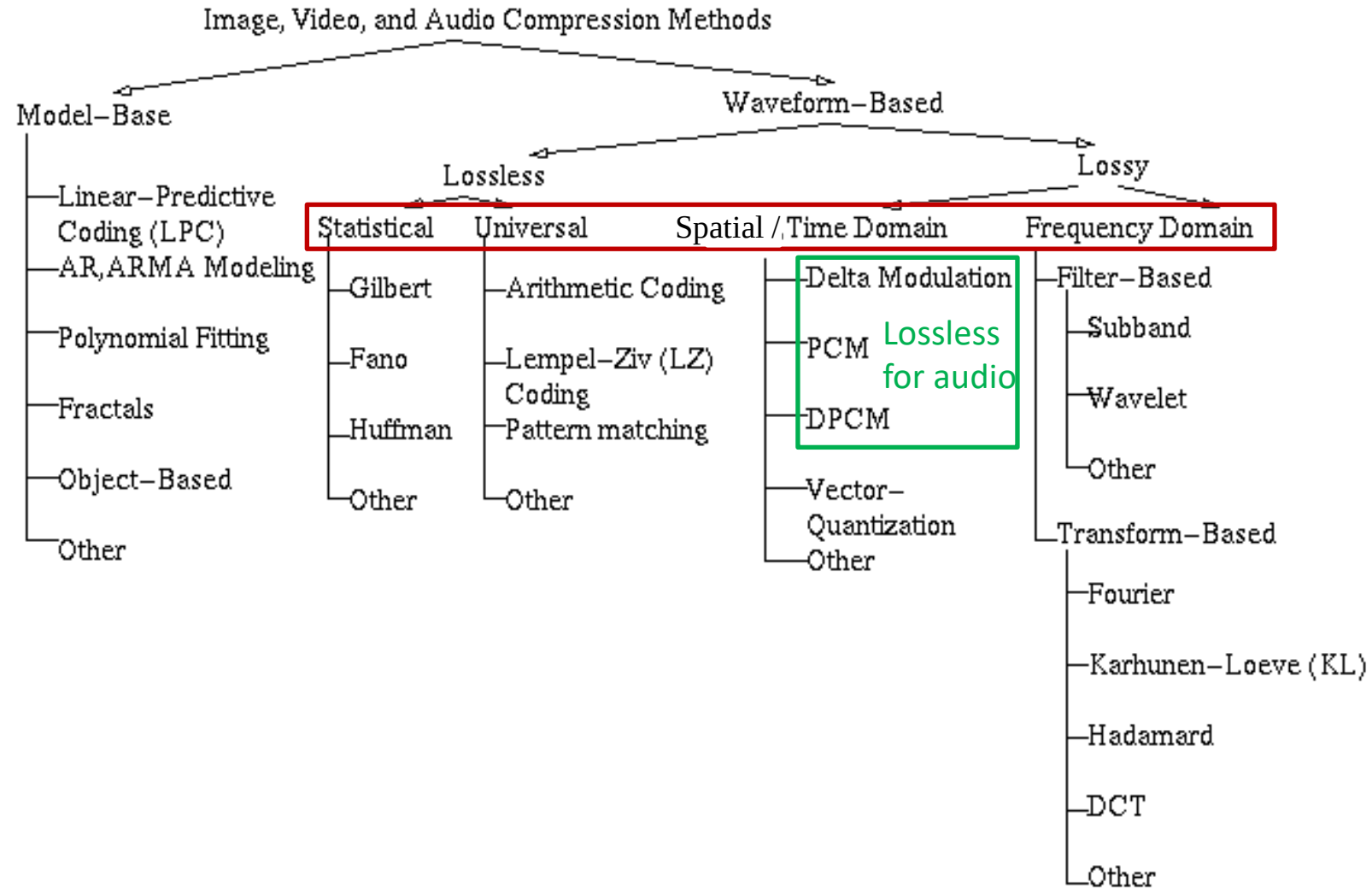
- A codec is a device or computer program which encodes or decodes a digital data stream or signal
- Typically a pair of algorithms for encoding and decoding
- Codec is a blend of the words en**code**r – **de**code
- An audio codec converts analog audio signals into digital signals for transmission or encodes them for storage. A receiving device converts the digital signals back to analog form using an audio decoder for playback.
- A codec may also **compress** the data to reduce transmission bandwidth or storage space

Compression Classification

- **Lossless** Compression
 - decompressed data is identical to original data.
- **Lossy** compression (irreversible compression)
 - uses inexact approximations and partial data discarding to represent the content



Compression Taxonomy



Audio Codecs

CODECS

VIDEO

H.264 (AVC)
H.265 (HEVC)
VP9
AV1

AUDIO

MP3
AAC

AAC

Advanced Audio Coding



Widely supported



More efficient than MP3



Limit on audio channels

Motion Picture Experts Group (MPEG)

- General Information about MPEG
- MPEG Objectives and Standards
- Audio Codecs:
 - MPEG-1 / MPEG-2 Audio Layer 3 (MP3)
 - MPEG-2 / MPEG-4 Advanced Audio Coding (AAC)

Motion Picture Experts Group (MPEG)

About 350 members from 200 industries, universities, and research institutions from 20 countries

ISO/IEC JTC 1/SC 29 **Coding of audio, picture, multimedia and hypermedia information** (since 1991) is a standardization subcommittee (7 working groups, 3 advisory groups)

- SC 29/AG 2: MPEG Technical Coordination
- SC 29/AG 3: MPEG Liaison and Communication
- SC 29/AG 5: MPEG Visual Quality Assessment
- SC 29/WG 2: MPEG Technical requirements
- SC 29/WG 3: MPEG Systems
- SC 29/WG 4: MPEG Video Coding
- SC 29/WG 5: MPEG Joint Video Coding Team(s) with ITU-T SG 16
- SC 29/WG 6: MPEG Audio coding
- SC 29/WG 7: MPEG 3D Graphics coding
- SC 29/WG 8: MPEG Genomic coding

MPEG Objectives

- Deliver acceptable video quality at compressed data rates (for MPEG-1 between 1.0 and 1.5 Mbps)
- Support either **symmetric** or **asymmetric** compress / decompress applications
 - symmetric = compression/decompression takes same time
- If compression takes it into account, **random-access** playback is possible
- If compression takes it into account, fast-forward, fast-reverse or normal reverse playback modes are available
- Audio/video **synchronization** will be maintained

MPEG Objectives

- Catastrophic behavior in the presence of data errors should be avoidable
- If required, compression/decompression delay can be controlled
- Editability should be available if required by applications
- Processing requirements should allow the development of low-cost chipsets which are capable of encoding in real-time

MPEG Standards

- Standard specifies audio, video and system layers
- Many standards defined
- MPEG-1 targeted at low data rates
(VHS quality at 1.5 Mbits/ sec) [1992]
- MPEG-2 targeted higher quality, hence higher data rates
(studio quality up to 15 Mbits/sec) [1994]
- MPEG-4 targeted at very low bit rates (<64 kbs) with small images [1998]

Acronym for a group of standards ⇅	Title ⇅	ISO/IEC standards ⇅	First public release date (First edition) ⇅
MPEG-1	Coding of moving pictures and associated audio for digital storage media. Commonly limited to about 1.5 Mbit/s although specification is capable of much higher bit rates Audio Layer 3 = MP3	ISO/IEC 11172	1993
MPEG-2	Generic coding of moving pictures and associated audio information Audio Layer 3 = MP3	ISO/IEC 13818	1995
MPEG-3			
MPEG-4	Coding of audio-visual objects H.264 = MPEG-4/Part 10 = MPEG-4 AVC	ISO/IEC 14496	1999
MPEG-7	Multimedia content description interface AAC = MPEG-4 Part 3, MPEG-4 Audio or ISO/IEC 14496-3:1999	ISO/IEC 15938	2002
MPEG-21	Multimedia framework (MPEG-21)	ISO/IEC 21000	2001
MPEG-A	Multimedia application format (MPEG-A)	ISO/IEC 23000	2007
MPEG-B	MPEG systems technologies	ISO/IEC 23001	2006
MPEG-C	MPEG video technologies	ISO/IEC 23002	2006
MPEG-D	MPEG audio technologies	ISO/IEC 23003	2007
MPEG-E	Multimedia Middleware	ISO/IEC 23004	2007
MPEG-G	Genomic Information Representation	ISO/IEC 23092	2019
(none)	Supplemental media technologies	ISO/IEC 29116	2008
MPEG-V	Media context and control	ISO/IEC 23005 ^[42]	2011
MPEG-M	MPEG extensible middleware (MXM)	ISO/IEC 23006 ^[47]	2010
MPEG-U	Rich media user interfaces	ISO/IEC 23007 ^[49]	2010
MPEG-H	High Efficiency Coding and Media Delivery in Heterogeneous Environments H.265 = High Efficiency Video Coding (HEVC), MPEG-H Part 2	ISO/IEC 23008 ^[54]	2013
MPEG-DASH	Information technology – DASH Dynamic adaptive streaming over HTTP (DASH)	ISO/IEC 23009	2012
MPEG-I	Coded Representation of Immersive Media H.266 = Versatile Video Coding (VVC), ISO/IEC 23090-3, MPEG-I Part 3, Future Video Coding (FVC)	ISO/IEC 23090	2020

MPEG-H 3D Audio, Audio Objects



MPEG General Information

- MPEG standard defines audio, video coding and system data streams with synchronization
- MPEG considers explicitly functionalities of other standards, e.g. it uses JPEG
- MPEG stream provides information on
 - aspect ratio (e.g., 1:1, 4:3, 16:9)
 - refresh frequencies (8 frequencies) encoded:
23.976 Hz, 24 Hz, 25 Hz, 29.97 Hz, 30 Hz, 50 Hz, 59.94 Hz and 60 Hz

```
▼ Stream 0
  Codec: H264 - MPEG-4 AVC (part 10) (avc1)
  Type: Video
  Video resolution: 1280x720
  Buffer dimensions: 1280x720
  Frame rate: 23.976024
  Decoded format:
  Orientation: Top left
  Chroma location: Left
▼ Stream 1
  Codec: MPEG AAC Audio (mp4a)
  Type: Audio
  Channels: Stereo
  Sample rate: 48000 Hz
  Bits per sample: 32
```

MPEG-1 General Information

- Audio compressor uses subband coder with psychoacoustic model to reduce bit rate
- Video compressor uses block transform coder with motion- compensated inter-frame coding
- Audio and video compression designed to operate at CD-ROM speeds (video—1.150 Mbits/sec, audio—0.256 Mbits/sec, system 0.094 Mbits/sec)

MPEG-1 Audio Standard

- Precision: 16 Bits, sampling frequencies 32kHz, 44.1kHz, 48kHz
- Audio coding: single channel, 2 independent channels, 1 stereo channel
- 3 compression methods: Layer 1, Layer 2, Layer 3 (hierarchical decoding)
 - Layer 3: decoder is accepting Layer 2 and Layer 1 (32kbps- 320kbps, target 64 kbps)
 - Layer 2: decoder is accepting Layer 1 (32 kbps-384kbps, target 128 kbps)
 - Layer 1: 32 kbps-448kbps, target 192 kbps

MPEG Audio

- *MPEG-1/2 Layer-3 is called MP3*
- Exponential growth of MPEG coded audio material on the internet since 1995
- “mp3” the most searched for term in early 1999
- Everybody is using MP3, not many know details of MPEG audio coding
- Compared to CD-quality digital audio, **MP3 compression can commonly achieve a 75 to 95% reduction in size**. For example, an MP3 encoded at a constant bitrate of 128 kbit/s would result in a file **approximately 9% of the size of the original CD audio**.

Why MPEG-1/2 Layer-3?

- Open standard
 - Specified in 1993
 - specification available to everybody (patents expired 2017)
 - no single company “owns” the standard
 - public example source code available to implementers
 - format well defined
- Availability of encoders and decoders
- Widespread use and support; computers fast enough for encoding & decoding

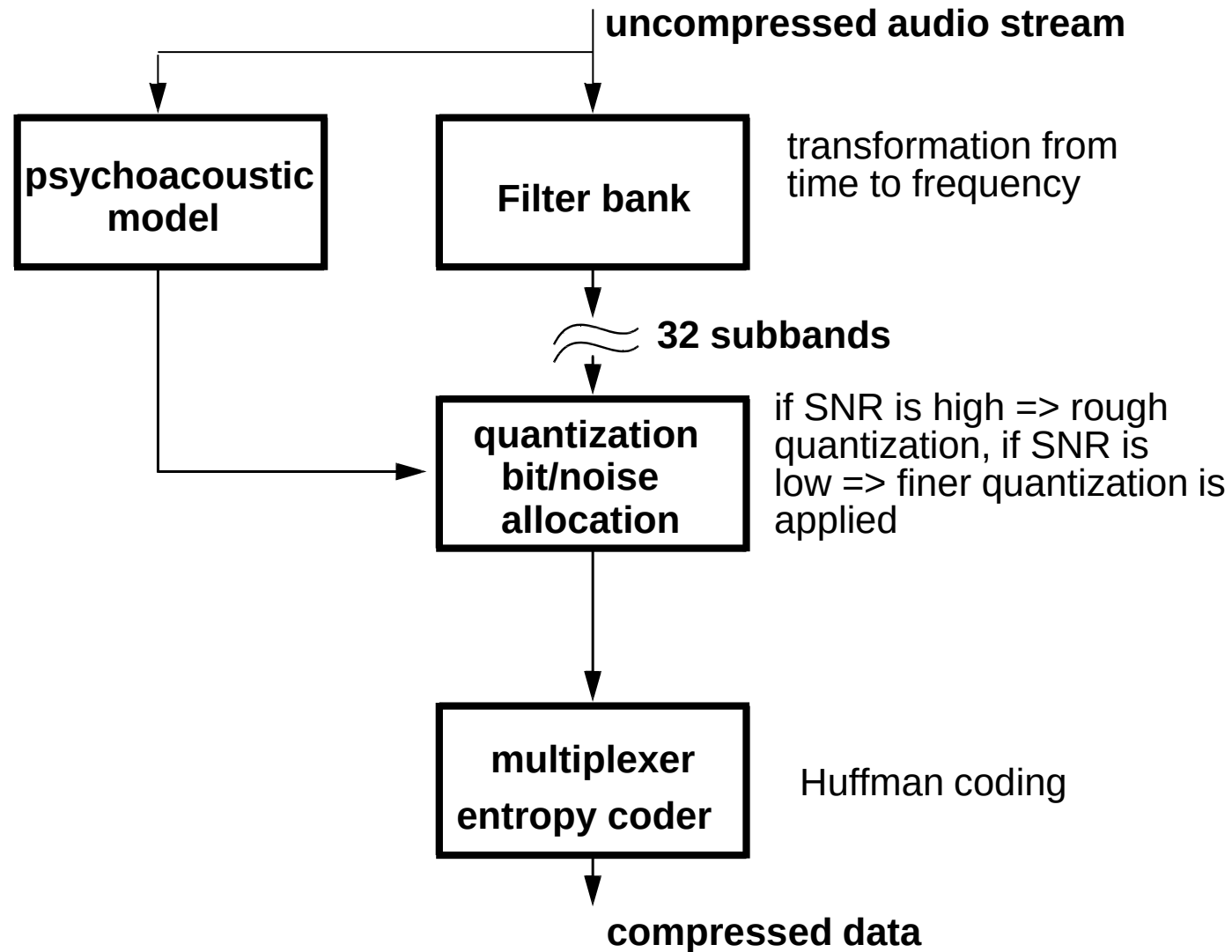
Anecdote:

A hacker named SoloH discovered the source code of the "dist10" MPEG reference implementation shortly after the release on the servers of the University of Erlangen. He developed a higher-quality version and spread it on the internet. This code started the widespread CD ripping and digital music distribution as MP3 over the internet.

General Approach of MPEG-1/2 Layer-3

- Encoder is not precisely specified but building blocks are described
 - Implementers of the standard were supposed to devise their own algorithms suitable for removing parts of the information from the audio input.
 - Many different MP3 encoders became available, each producing files of differing quality.
- Decoding is carefully defined in the standard. Most decoders are "bitstream compliant"
 - Only computational performance of decoders differs (depending how efficiently implemented)

MPEG Audio Standard (Encoding)



High Quality Audio Coding

- MPEG-1 Layer-3 has been specified 1993
- since then, research on perceptual audio coding has progressed and codes with better compression efficiency became available: MPEG-2 Advanced Audio Coding (AAC) and other proprietary compression systems.
- Basic task is to compress audio data in a way that
 - the compression is as efficient as possible (file size small)
 - the reconstructed (decoded) audio sounds exactly (or as close as possible) to the original audio data
 - requires low complexity (in software or inexpensive hardware)
 - offers flexibility for different application scenarios.

Masking and MPEG Audio Compression

- General audio signal consists of many frequency components
 - each frequency component influences the hearing threshold curve
 - Audio signal varies with time
- Hearing threshold different at each point in time
- **Psychoacoustic model**
 - Calculates current hearing threshold curve in each case
- **Variable quantization**
 - Frequency components quantized so that quantization noise just below current hearing threshold curve
 - Where masking is used, coarse quantization, small word lengths

A Basic Perceptual Audio Encoder

Consists of the following building blocks:

1. Filter bank—is used to decompose the input signal into subsampled spectral components (time/frequency domain)
2. Perceptual model—using either the domain input signal and/or the output of the analysis filter bank, an estimate of the actual masking threshold is computed by rules from psychoacoustics
3. Quantization and coding—the spectral components are quantized and coded; noise introduced should be kept below threshold
4. Lossless Huffman Encoding of bitstream

A Basic Perceptual Audio Coder

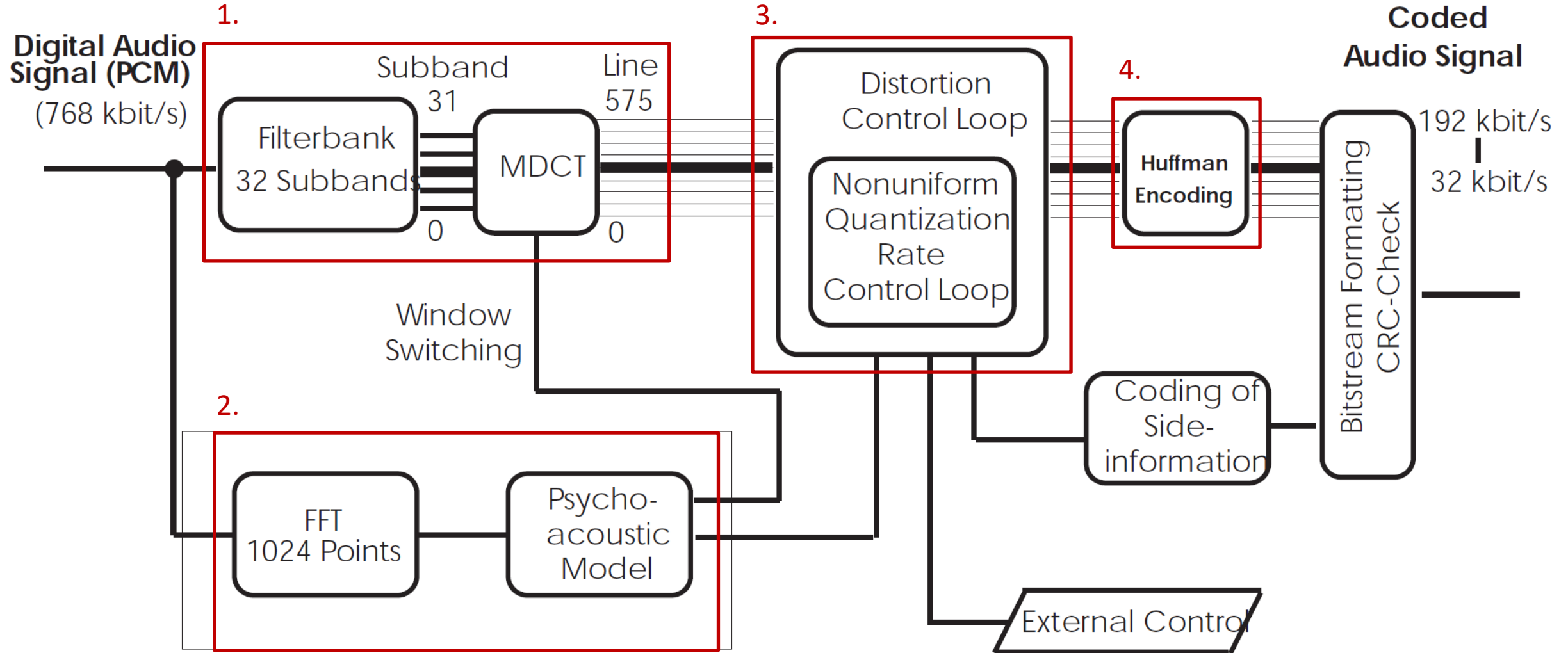


Figure 2: Block diagram of an MPEG-1 Layer-3 encoder.

Layer-3 Encoding Algorithm

Basic building blocks

- **Filterbank**: Actually 2 cascading filterbanks:
 - Polyphase filterbank (as in Layer-1 and 2); Cut off frequencies $<16\text{Hz}$, $>16\text{kHz}$
 - Additional Modified DCT (MDCT); the subdivision of each polyphase frequency band into **18 finer subbands** increases the potential for redundancy removal;
- **Perceptual Model**
 - Mainly determines the quality of a given encoder implementation; uses either a separate filterbank or combines the calculation of energy values (for masking calculations) and main filter bank

Layer-3 Encoding Algorithm

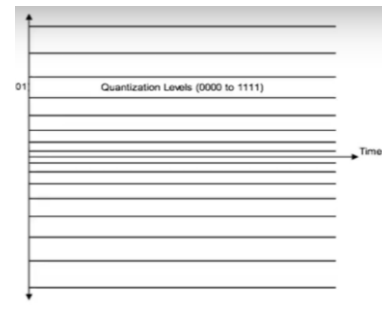
- Perceptual Model (continued)
 - Output of the perceptual model consists of values for the masking threshold or allowed noise for each coder partition (roughly equivalent to the critical bands of human hearing)
- **Quantization** and coding
 - two nested iteration loops (rate loop and noise control loop)
 - quantization is done via a power-law quantizer; larger values are automatically coded with less accuracy
 - quantized values are coded by **Huffman coding**

Layer-3 Encoding Algorithm

- Quantization and coding (continued)
 - outer iteration loop (noise control loop)

If the quantization noise in a given band is found to exceed the masking threshold (allowed noise) as supplied by the perceptual model, the **scale factor** for this band is adjusted to reduce the quantization noise.
(power-law quantizer = non-linear, can adjust quantization steps)
= **adapt quantization steps per band**
 - inner iteration loop (rate loop – modifies overall coder rate)

If number of bits resulting from coding operation exceeds the number of **bits available** to code a **block** of data, a larger quantization step is applied.
= **adjust global gain**



Huffman Encoding—Example

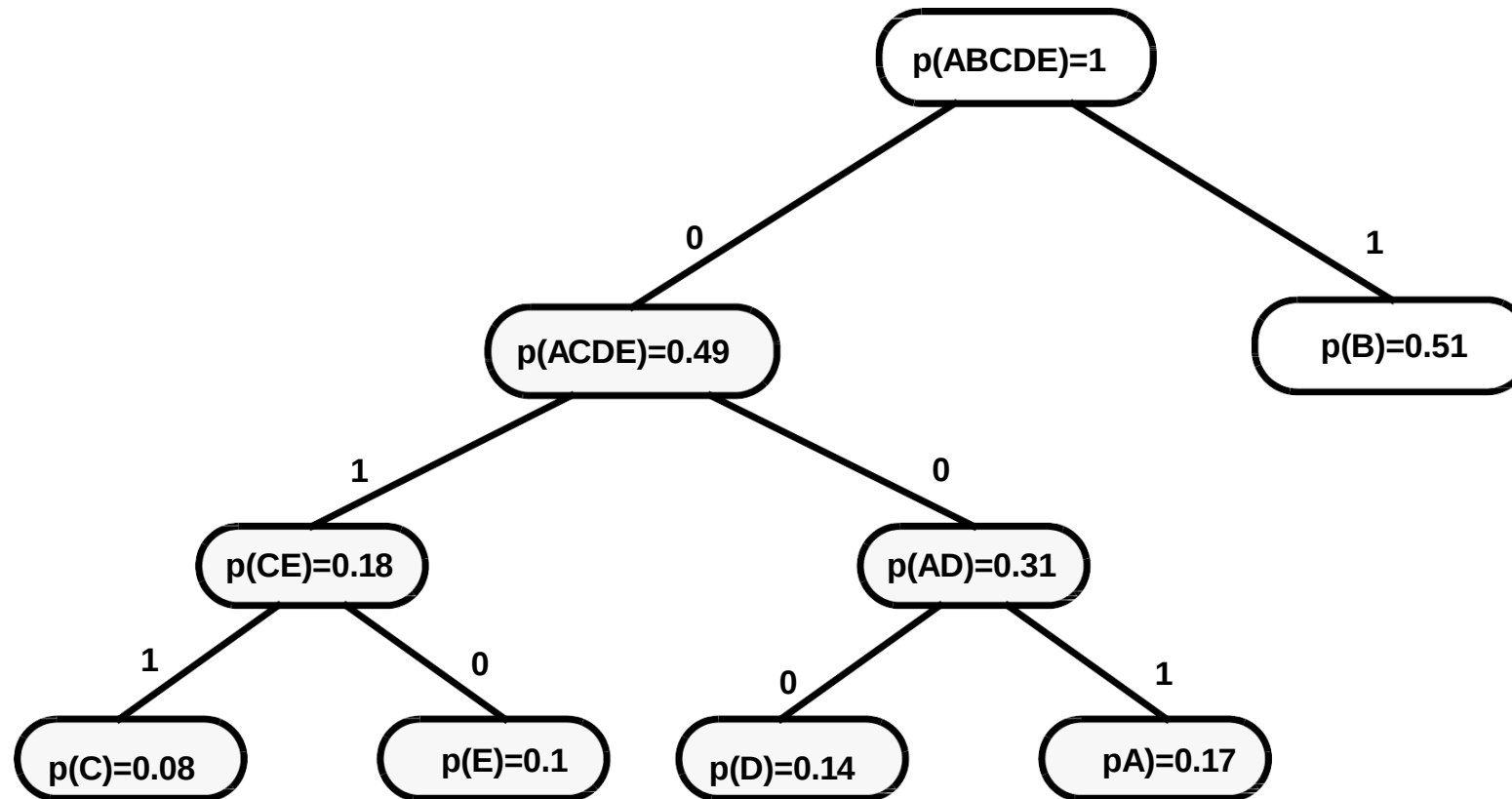
encode symbols according to tree

A	001
B	1
C	011
D	000
E	010

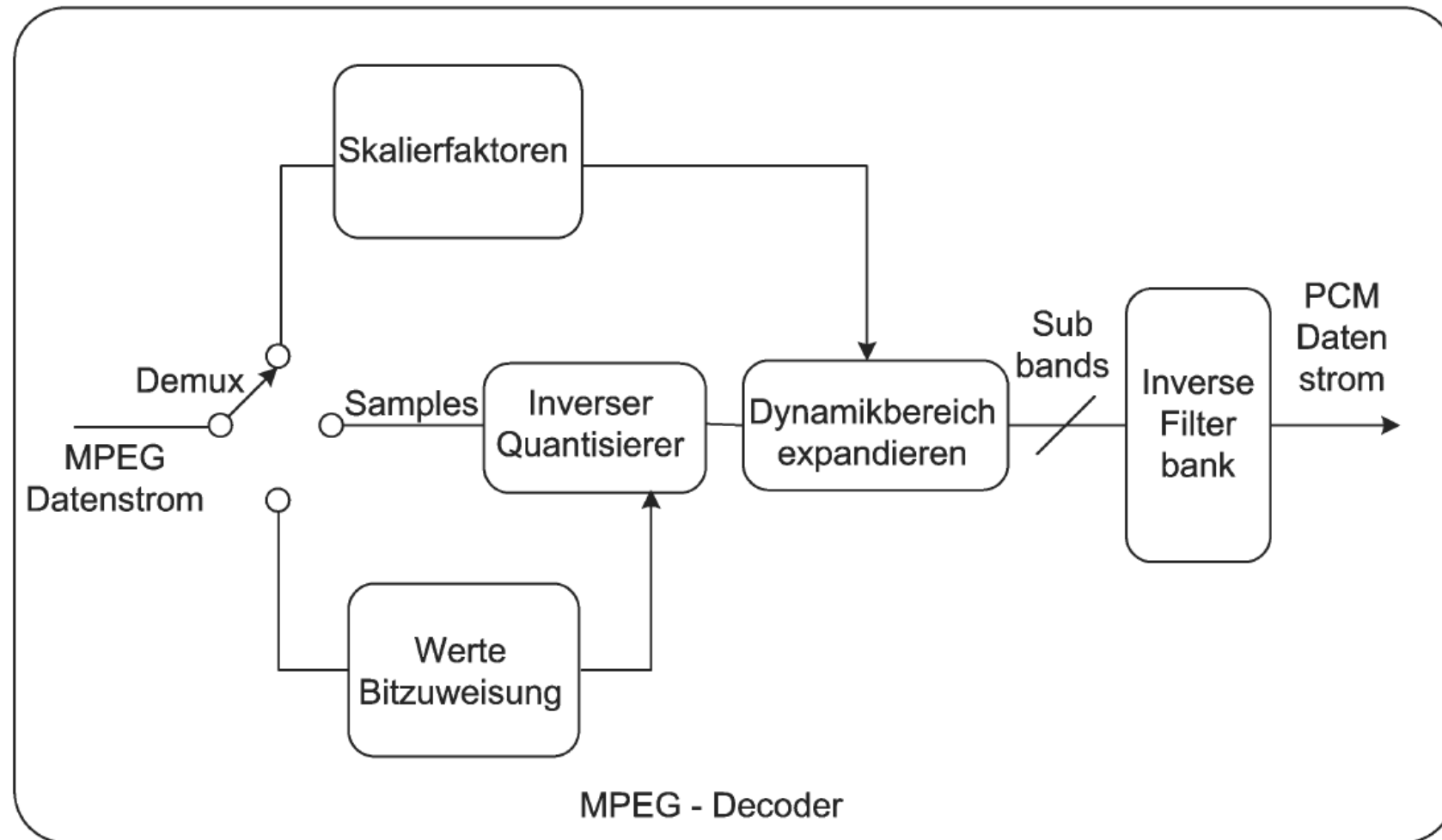
$p(C)=0.08$, $p(E)=0.1$, $p(D)=0.14$, $p(A)=0.17$, $p(B)=0.51$

Huffman Encoding—Example

last step: $p(ACDE)=0.49$, $p(B)=0.51$



Simple MPEG decoder



First Step: Split Audio File into “Frames”

- Split audio file into “frames”
- During encoding 576 time domain samples are taken & transformed to 576 frequency domain samples. If there is a transient 192 samples are taken.
 - This is done to limit the temporal spread of quantization noise accompanying the transient.
- The decoder expects a frame consisting of 384, 576, or 1152 samples
 - Typically in MP3 each "frame" has a length of 1152 samples = 0.026 sec (26,122449 ms)
- The following processing happens in each frame individually - with some overlap with previous/next frame to minimize temporal effects

MP3 Algorithm – Basic Steps (Simplified)

```
algorithm MPEG-1_audio
```

```
/*Input: An audio file in the time domain
```

```
Output: The same audio file, compressed*/
```

```
{
```

```
  Divide the audio file into frames
```

```
  For each frame {
```

```
    By applying a bank of filters, separate the signal into frequency bands.
```

```
    For each frequency band {
```

```
      Perform a Fourier transform to analyze the band's frequency spectrum
```

```
      Analyze the influence of tonal and nontonal elements (i.e., transients)
```

```
      Analyze how much the frequency band is influenced by neighboring bands
```

```
      Find the masking threshold and signal-to-mask ratio (SMR) for the band,  
      and determine the bit depth in the band accordingly
```

```
      Quantize the samples from the band using the determined bit depth
```

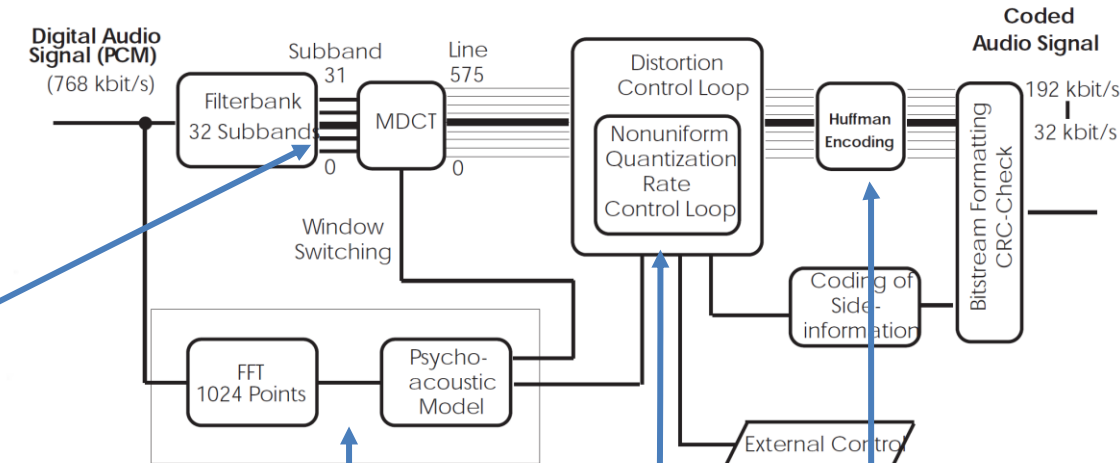
```
      Apply Huffman encoding (optional)
```

```
    }
```

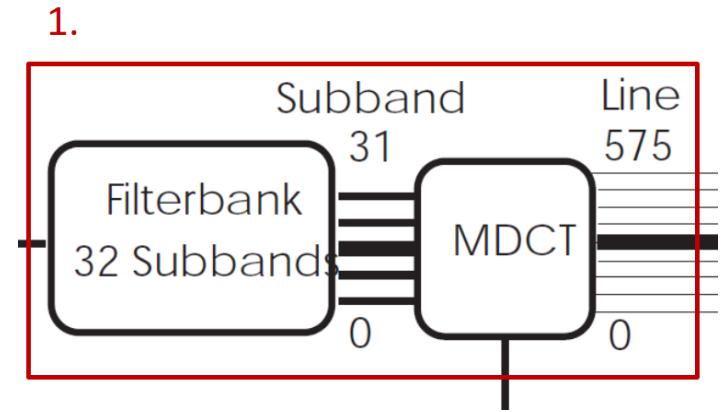
```
  Create a frame with a header and encoded samples from all bands
```

```
}
```

```
}
```



Subband coding

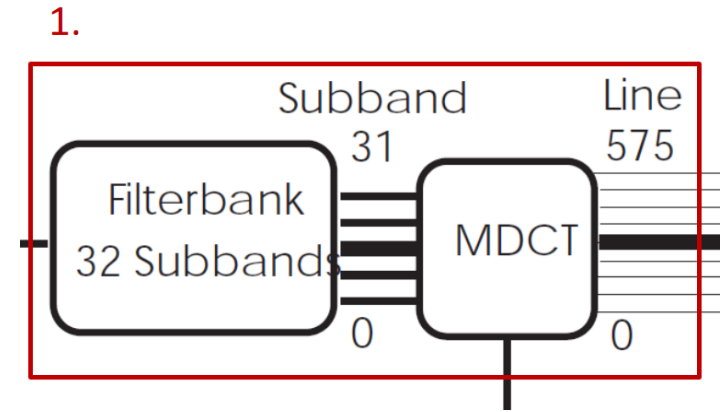


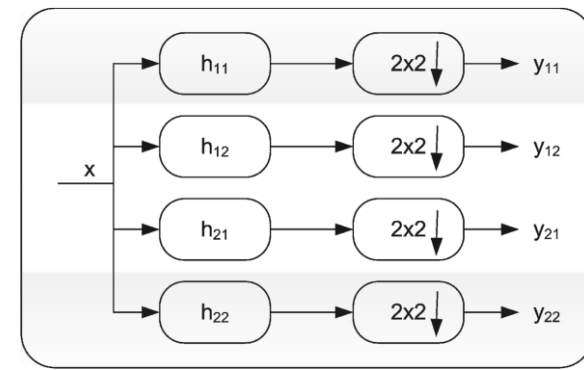
MPEG audio encodes frequency components

- Filter bank
 - divides audio signal into frequency bands (subbands)
- Ideal filter bank
 - Subbands identical to frequency groups of human hearing

MPEG Polyphase Filter Bank

- Input: Digital PCM Audio Signal
- MPEG Polyphase Filter Bank
 - Audio signal is divided into 32 subbands
 - Each subband has equal size/bandwidth
- Advantage
 - simple structure
- Disadvantage
 - lower subbands span over several frequency groups
 - solved in MPEG-1/Layer 3 by MDCT, modeling human critical bands more closely
 - overlapping subbands lead to information loss





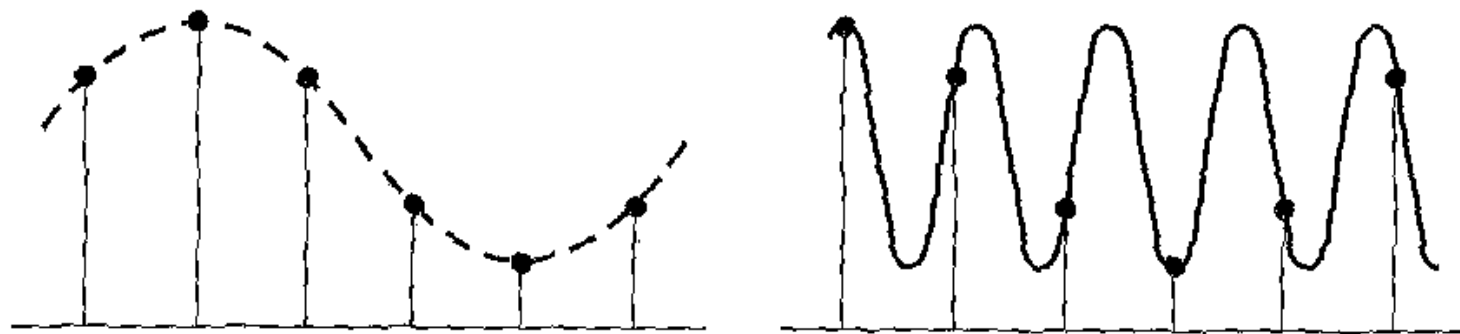
- Example filter bank with 4 subbands
- Input - 1 x PCM: frequency range 0 - 20kHz, sampling rate 48 kHz; word length 16 bit per sample
- Output - 4 x PCM
 - Subband 1: 0Hz - 5kHz; sampling rate 12kHz (!); 16 bit
 - Subband 2: 5kHz -10 kHz; sampling rate 12kHz (!); 16 bit
 - Subband 3: 10kHz - 15 kHz; sampling rate 12kHz (!); 16 bit
 - Subband 4: 15 kHz - 20 kHz; sampling rate 12kHz (!); 16 bit
- Seems to be a violation of the Nyquist-Shannon theorem BUT (see [1])
 - IF a **band does not start at zero frequency** but at some higher value and can be shifted by a linear translation, smaller sampling rate – **at least twice the width of the non-zero frequency interval** - is required.
 - For example, in order to sample the FM radio signals in the frequency range of 100–102 MHz, it is not necessary to sample at 204 MHz (twice the upper frequency), but rather it is sufficient to sample at 4 MHz (twice the width of the frequency interval).

[1]

https://en.wikipedia.org/wiki/Nyquist%E2%80%93Shannon_sampling_theorem#Sampling_of_non-baseband_signals

Filter Bank - Critical Sampling

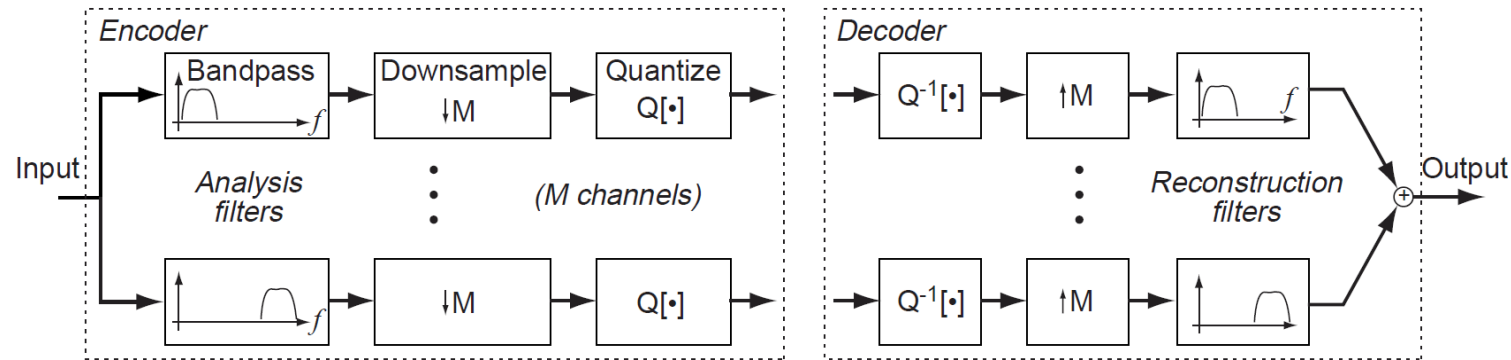
- Typically 32 subbands in MP3
- Critical sampling of digital filter banks
 - Sampling rate in subbands reciprocal to the number of subbands
 - Reduced sampling rate in subbands is called critical sampling rate
- Critical sampling of subbands ensures that **filter bank has no influence on data volume**
 - More channels are generated, but their sampling frequencies are correspondingly lower.



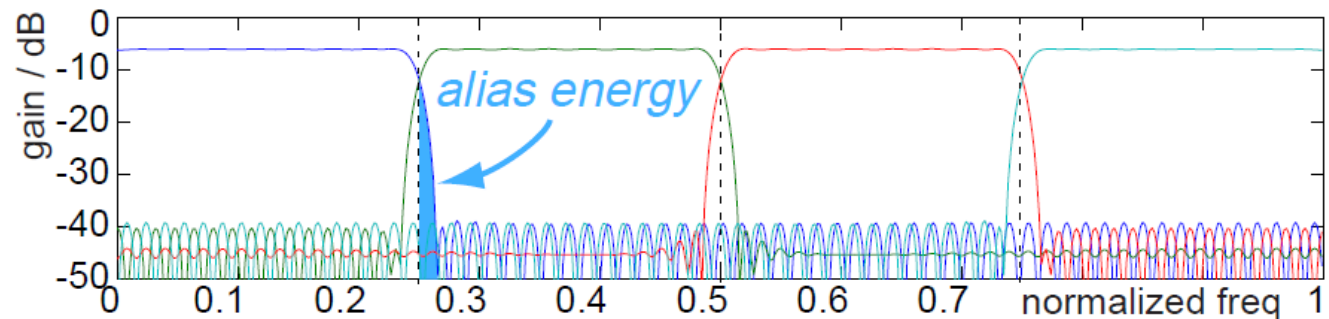
- Left: Samples / signal without specification of frequency range
- Right: Samples / signal with indication of frequency range (right)

Subband Coding

- Idea: Quantize separately in separate bands
 - Quantization Noise stays within band, gets masked



- 'Critical sampling' $\rightarrow 1/M$ of spectrum per band
 - some aliasing inevitable



- Trick is to cancel with alias of adjacent band
 - $\rightarrow$ 'quadrature-mirror' filters / MDCT

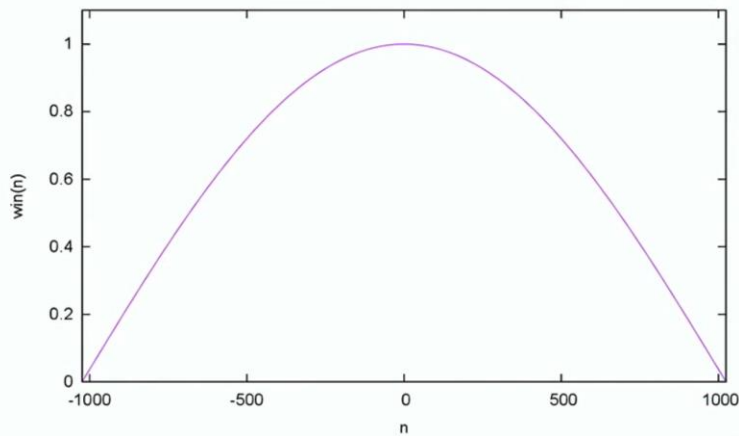
Subbands in MPEG Standard

- MPEG-1/Layer 1 and Layer 2 - 32 Subband Filter Bank
- In MPEG-1/Layer 3 (MP3) a Hybrid filter bank is used
 - Modified Discrete Cosine Transform (MDCT) performs transformation into frequency domain
- In MPEG-2 - AAC
 - Lossless MDCT - frequency decomposition corresponds to 1024 subbands

Modified Discrete Cosine Transform (MDCT)

- Performs transformation of subbands into frequency domain
- Overlaps with neighboring frequency bands (but does not increase sample size)
- MDCT also used for avoiding temporal alias effects

Modified Discrete Cosine Transform (MDCT)

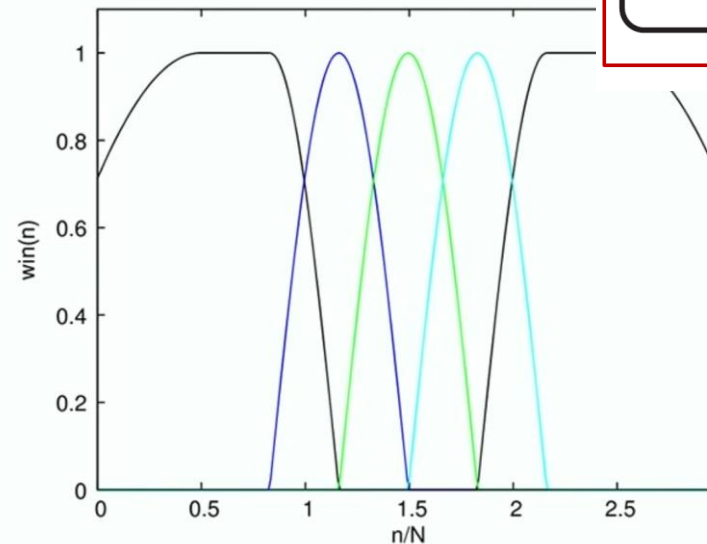


Sine window frequently used for MDCT

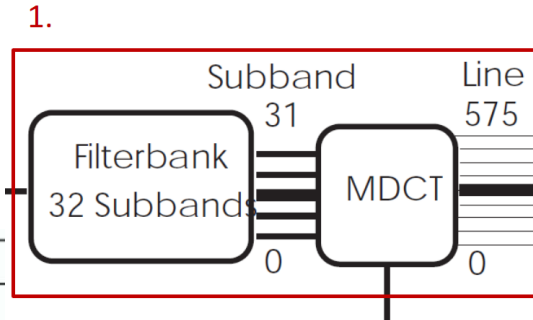


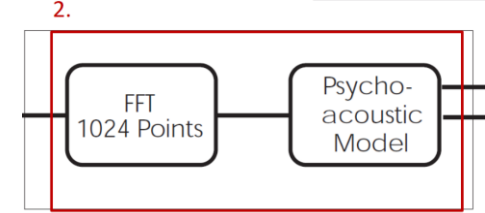
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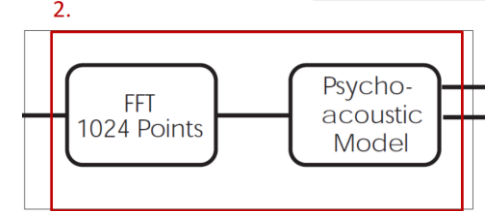
Adaptively switched MDCT windows





MPEG Psychoacoustic Model: Overview

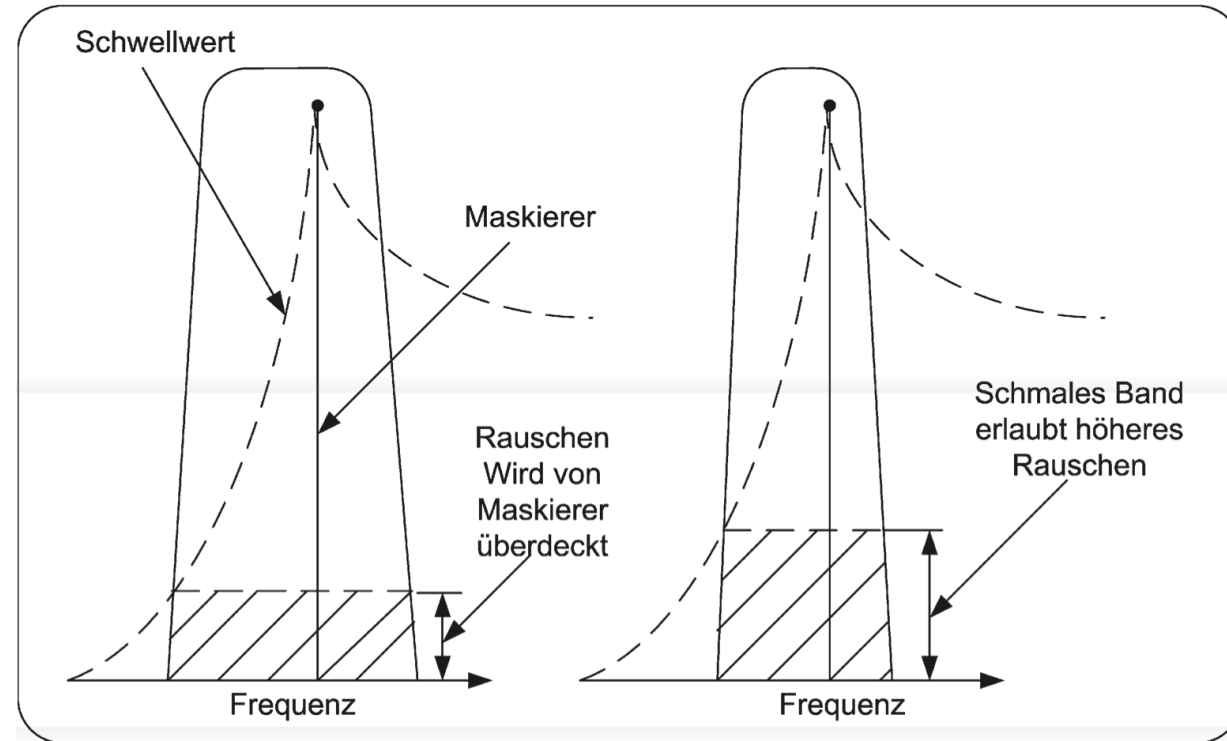
- First: Fast Fourier Transform into frequency domain
 - Psychoacoustic model operates on frequency spectrum
- Analyzes audio signal section by section for masking effects
 - Algorithms simulate human hearing
- Calculates masking threshold per subband
 - calculates maximum allowed quantization noise per subband
- Quality of the psychoacoustic model mainly responsible for overall quality of coding



Fast Fourier Transform

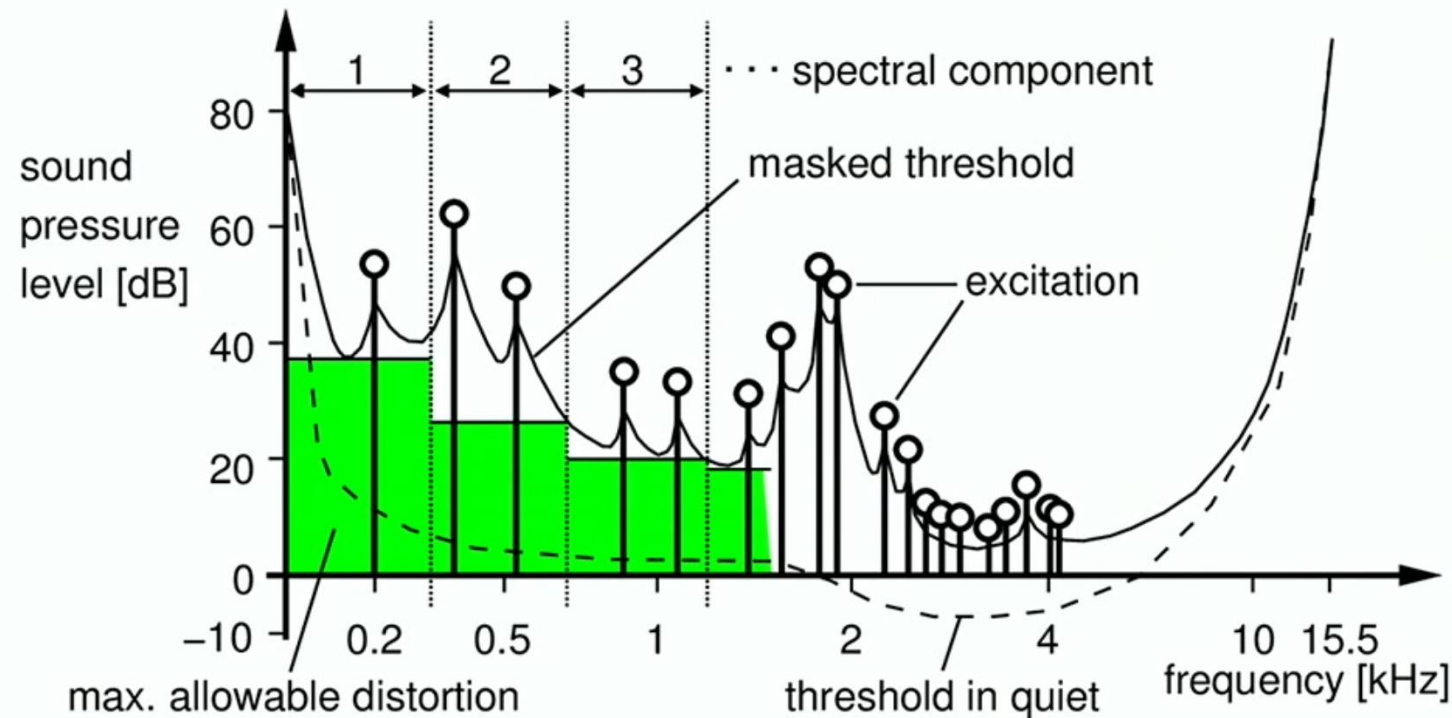
- For signal analysis FFT is particularly well suited
 - Provides fine frequency resolution of the signal
 - provides phase information of the frequency components (important for psychoacoustic model 2 (mentioned later))
 - Very accurate calculation of the masking thresholds
- ➡ high compression possible
- Technique for minimizing the computational effort
 - Bundling values in frequency groups
 - Frequency groups correspond to critical bandwidths of human hearing

Masking in Psychoacoustic Model

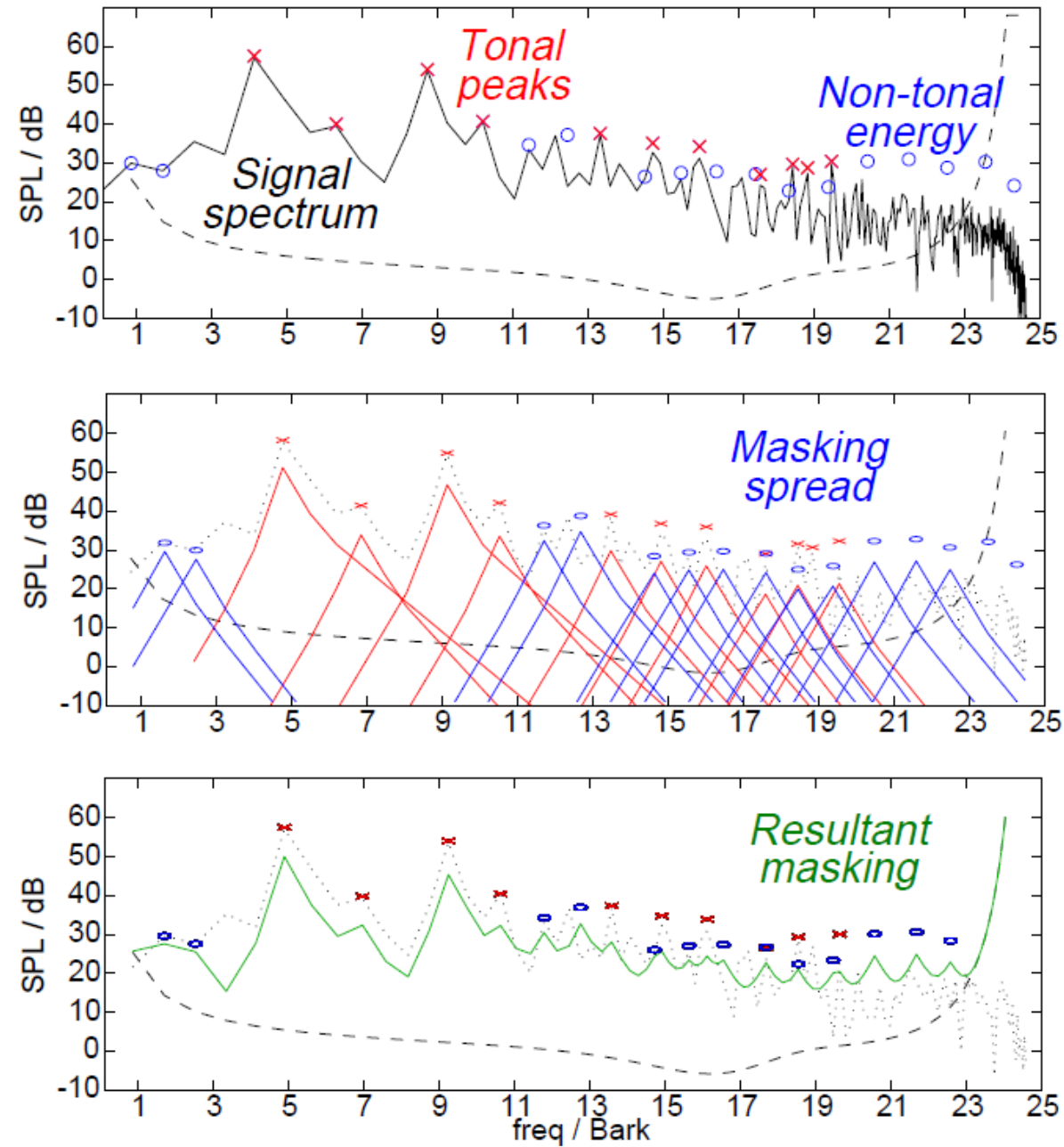


Masking in Psychoacoustic Model

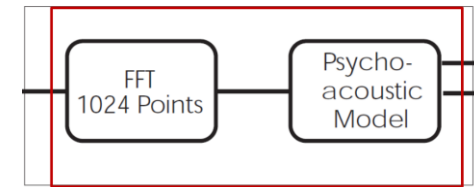
- For each subband: How high can quantization noise be, so that it is still masked?



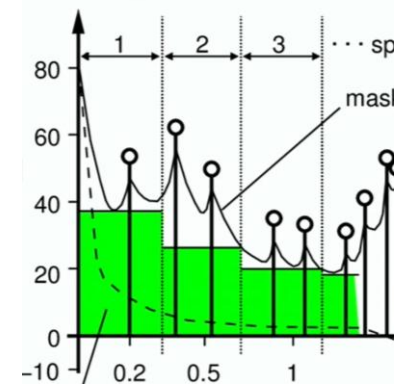
Spectral noise shaping in transform/subband coding

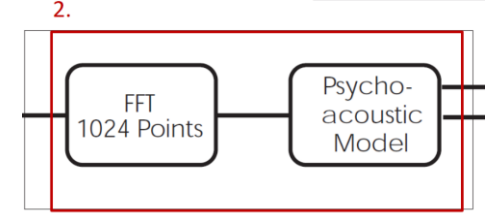


Masking in Psychoacoustic Model



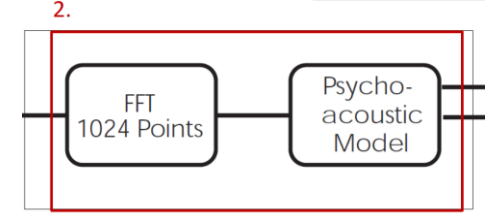
- Because of narrow subbands
 - high masking threshold for noise possible
 - frequency components require only few bits, which corresponds to high compression
- In general
 - the higher the subband number the narrower the subbands
 - the narrower the subbands the higher the compression





Tonal / Atonal Components

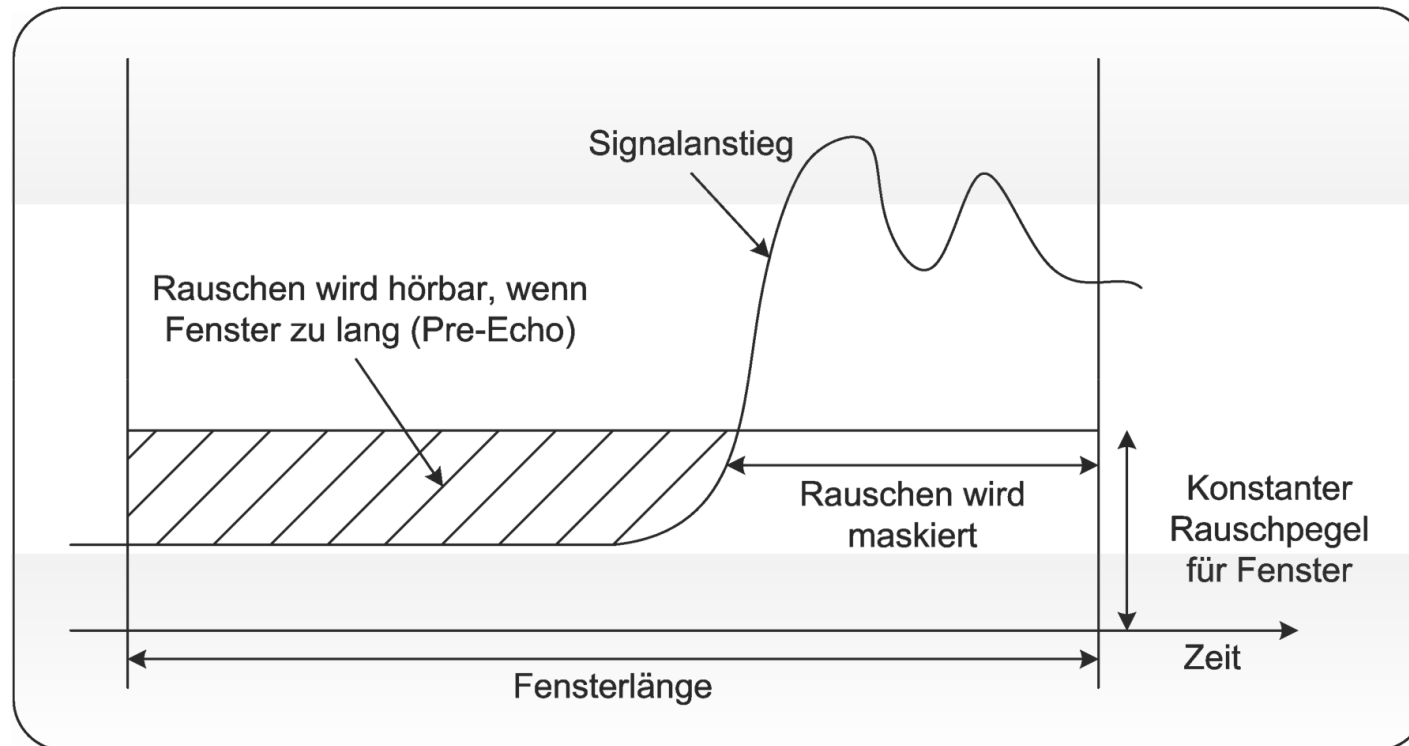
- Tonal and noise-like acoustic events have different masking properties
- Psychoacoustic model needs algorithms to distinguish between sound and noise
- Model 1
 - strong peaks in the frequency spectrum evaluated as tonal components
 - remaining parts interpreted as noises
- Model 2
 - more complex algorithms, considering phase, etc..



Result Psychoacoustic Model

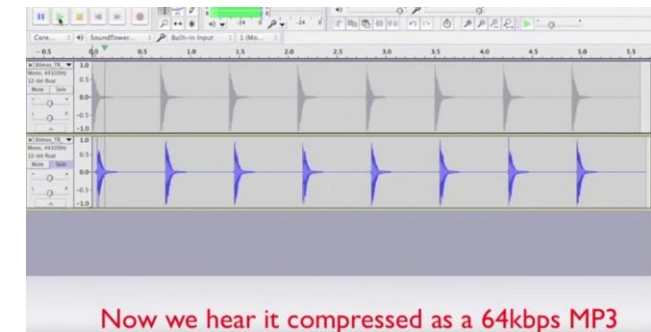
- Psychoacoustic model calculates hearing threshold curve
 - Noise below curve is not audible
 - Noise above curve is audible
 - for noise-free coding, audible noise level must not be exceeded at any point
- Calculation of the masking thresholds
 - for each subband the masking threshold is calculated
 - the smaller the subbands are, the better the fit of the step function to hearing threshold curve (high compression)

Compression with Steep Signal Rise – Temporal Issues



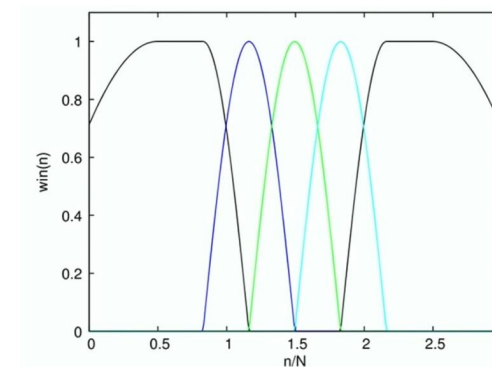
Compression with Steep Signal Rise – Pre-Echo

- Signal sections (windows) with steep signal slopes are very difficult to compress
 - Quantization noise equal over entire window length
 - Noise before signal rise audible as **pre-echo**
 - Quantization noise is unmasked in the silence before “attack”
- MPEG Strategies
 - Layer1/Layer2 - no artifact-free compression possible
 - MP3 - uses short window for this, pre-echo effect is reduced, but moderate compression, as less subbands are achievable with short window
 - AAC - short window, Temporal Noise Shaping (TNS) module, good compression



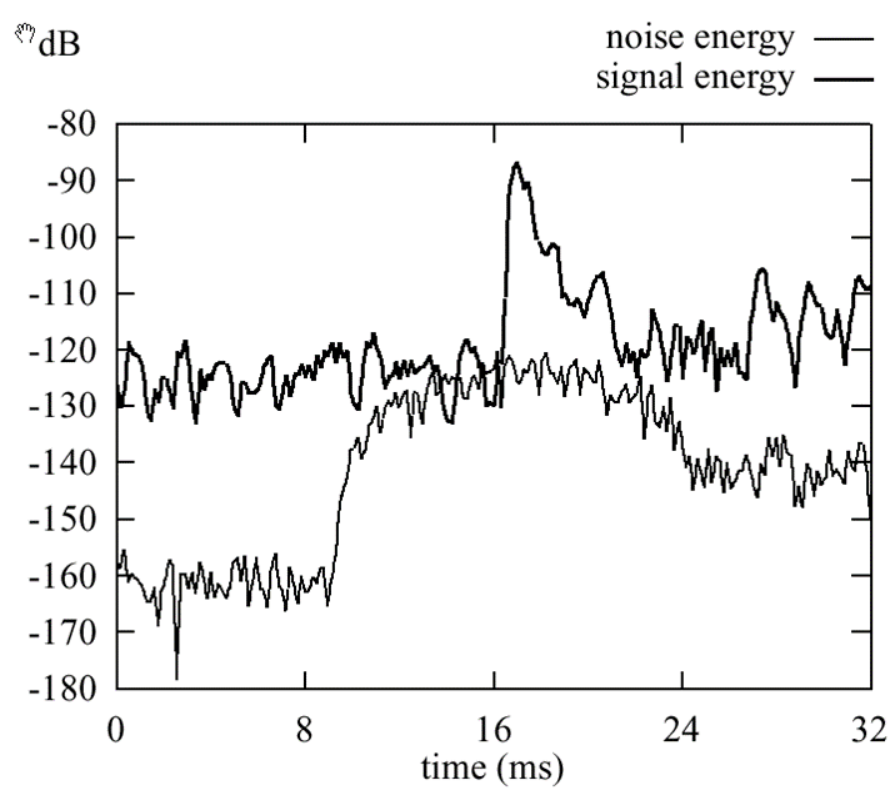
Pre-Echo

<https://www.youtube.com/watch?v=MtiRBFwKRs>

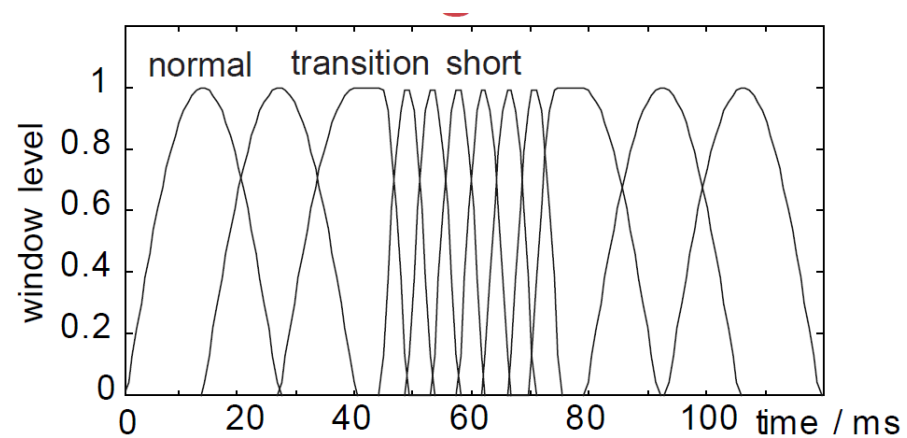


Adaptively switched MDCT windows

Pre-Echo



- Adaptive switching of time-window



Frequency Resolution / Time Window

High frequency resolution with long time window

+ : many subbands, high compression

- Subbands well adaptable to hearing threshold curve

- : Quantization noise not changeable for a long time

- bad with signal increases

Low frequency resolution with short time window

+ : Quantization noise changeable in short time intervals

- Noise can be well matched to time curve

- : Fewer subbands, moderate compression

A Basic Perceptual Audio Coder

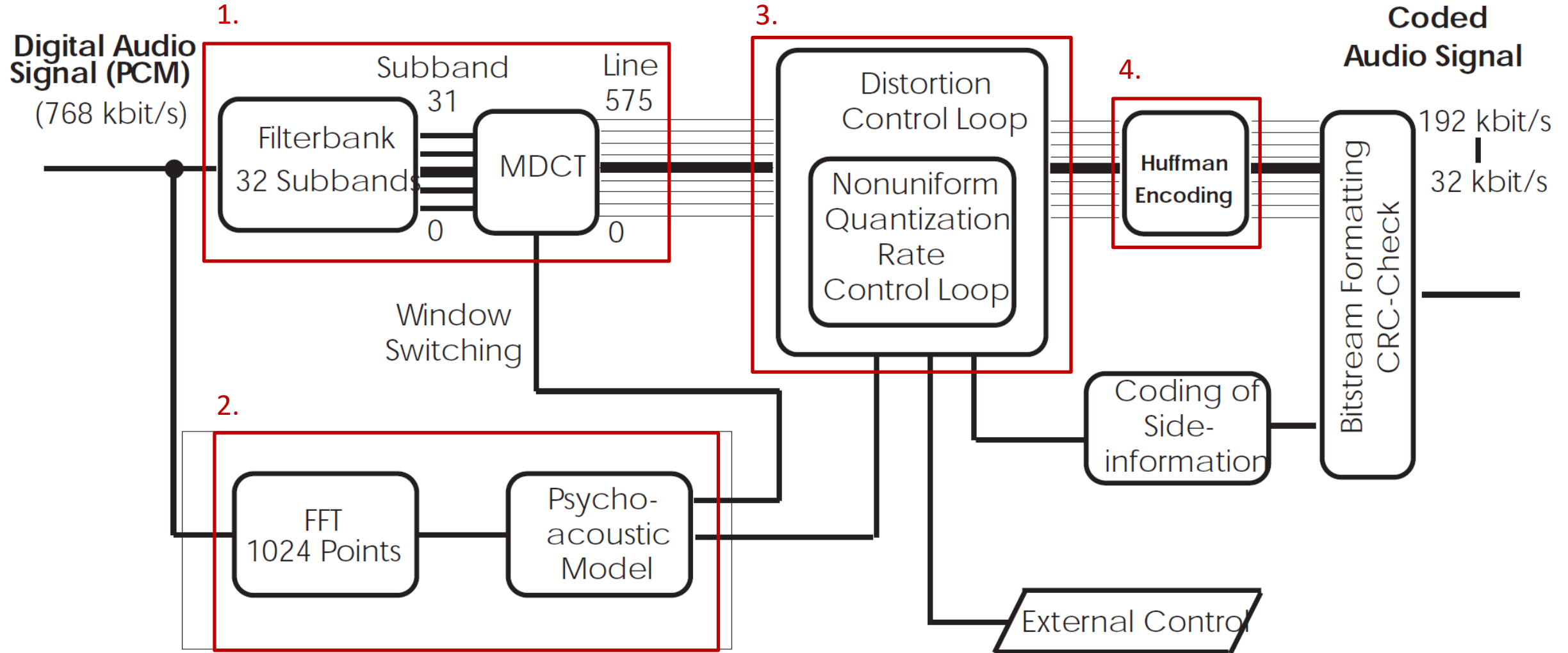
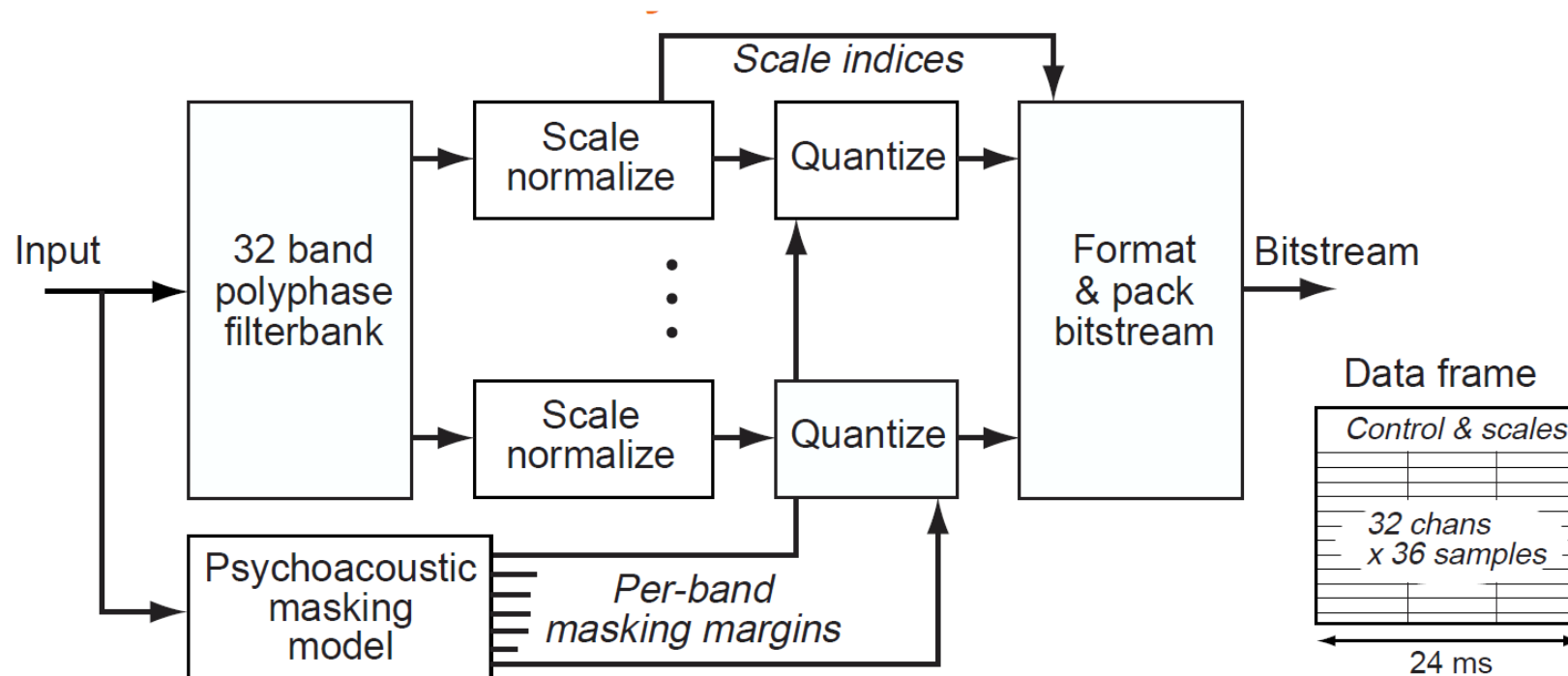


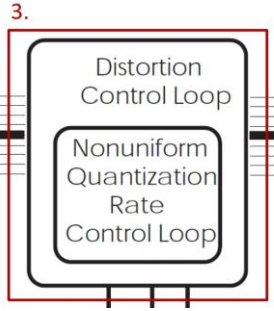
Figure 2: Block diagram of an MPEG-1 Layer-3 encoder.

General Quantization Approach

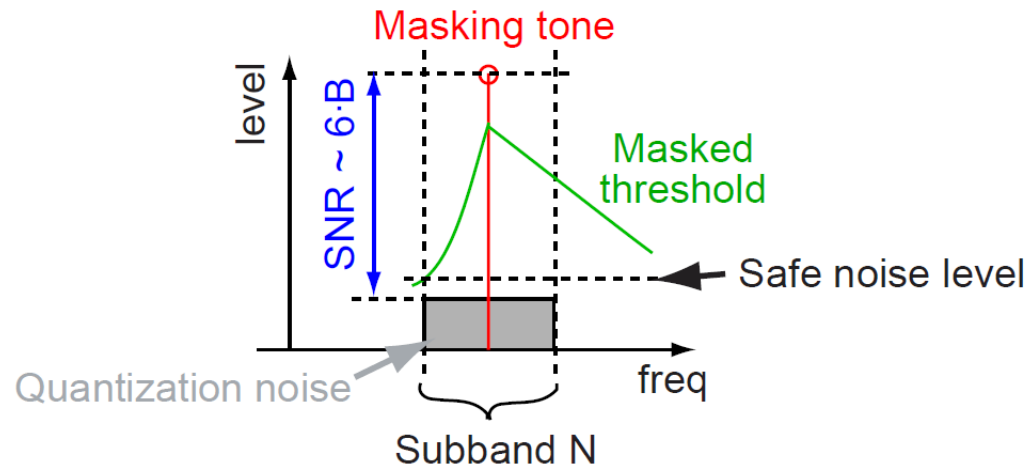
- Basic idea: Subband coding plus psychoacoustic masking model to choose dynamic Quantization levels in subbands



MPEG Bit allocation



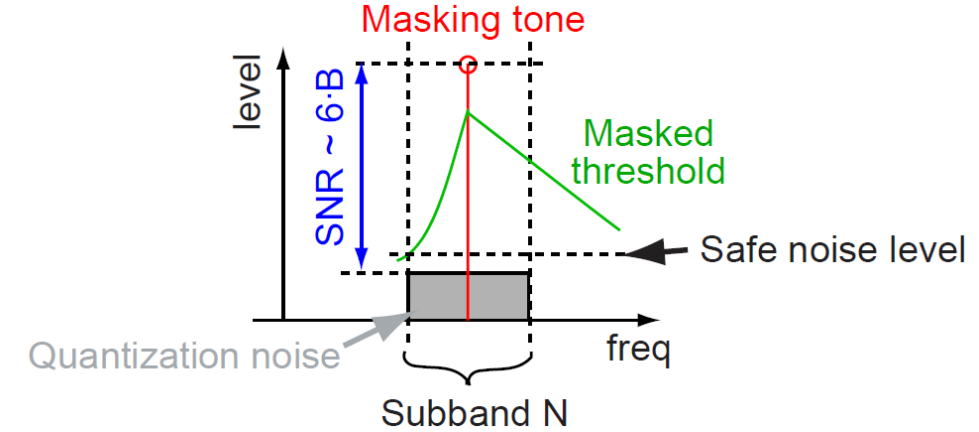
- **Result of psychoacoustic model is maximum tolerable noise per subband**



- safe noise level $\rightarrow$ required SNR $\rightarrow$ bits B

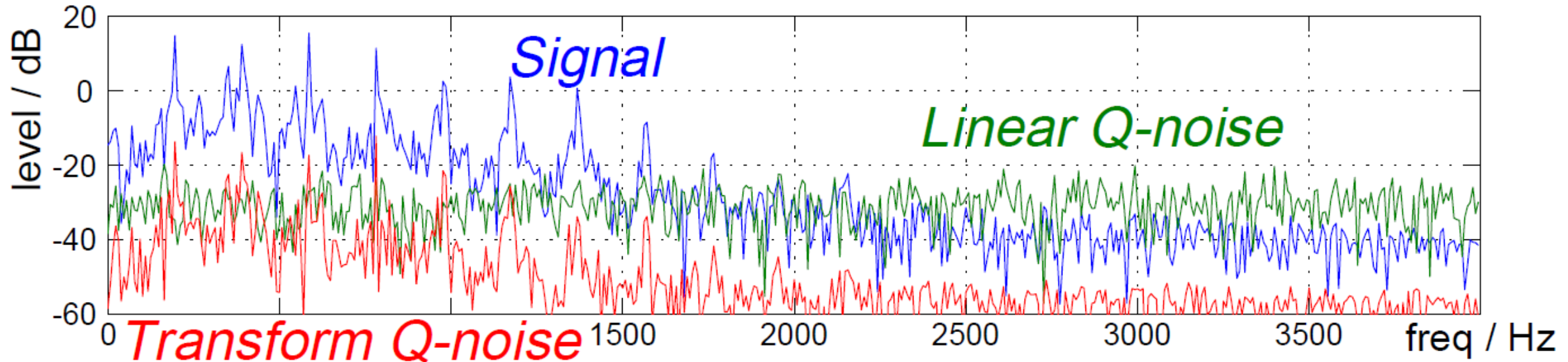
- **Bands with no signal above masking curve can be skipped entirely**
- **Bit allocation procedure (fixed bit rate):**
 - pick channel with worst noise-masker ratio
 - improve its quantization by one step
 - repeat while more bits available for this frame

Layer-3 Quantization Algorithm



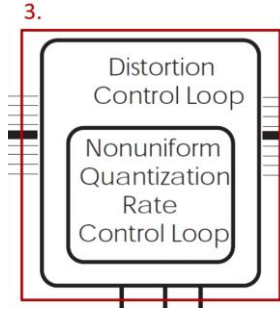
- **Quantization** and coding
 - quantization is done via a power-law quantizer; larger values are automatically coded with less accuracy
 - two nested iteration loops (rate loop and noise control loop)
- Outer iteration loop (noise control loop) – for each band
 - If the quantization noise in a given band is found to exceed the masking threshold (allowed noise) - supplied by the perceptual model - the scale factor for this band is adjusted to reduce the quantization noise.
- Inner iteration loop (rate loop – modifies overall coder rate) – for each frame
 - Check if Huffman coding is small enough
 - If number of bits resulting from coding operation exceeds the number of bits available to code a block of data, **adjust global gain** -> results in larger quantization step.

Example

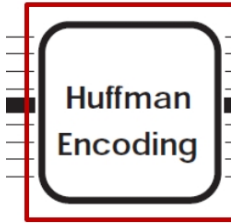


Comparison: Linear Quantization Noise over whole frequency band (green) compared to reduced quantization noise (in red) where noise is scaled in different frequency bands.

Bit assignment



- User can select desired MPEG bit rate
 - Psychoacoustic model calculates values so that no quantization noise is audible
 - at low data rates, available bits are usually not sufficient
 - Bit allocation algorithms
 - high frequencies are removed
 - Stereo will be spatially inaccurately resolved
 - Well-adjusted introduction of quantization noise
- Algorithms are left to software developers
- MPEG Audio standard offers tables as recommendation



Huffman Encoding

- Final quantized bands will be Huffman encoded = lossless encoding (the lossy part happened before)
- This is all still done in the frequency domain!

Huffman Encoding

- Statistical encoding, depends on **occurrence frequency of single characters or sequences** of data bytes
- Characters stored with their probabilities
- Length (number of bits) of the coded characters differs
- Shortest code assigned to the most frequently occurring character
- To determine Code, it is useful to construct a **binary tree**
 - leaves are characters to be encoded
 - nodes carry **occurrence probabilities** of the characters belonging to the subtree

Huffman Encoding — Algorithm

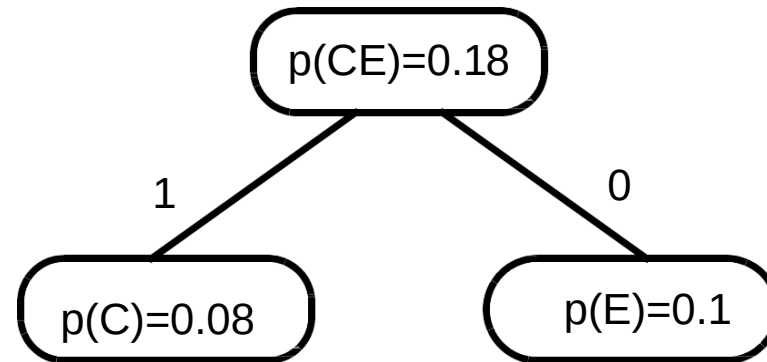
build a Huffman Tree:

$p(A)=0.17$, $p(B)=0.51$, $p(C)=0.08$, $p(D)=0.14$, $p(E)=0.1$

```
build ordered list of symbols (increasing probability)
do while list contains at least 2 elements
    construct tree using the first two elements in list
    add parent node for the union of these elements and compute probability
    mark edges by '0' and '1'
    delete the first 2 elements in list; insert parent into list
end
```

Huffman Encoding—Example

step 1: $p(C)=0.08$, $p(E)=0.1$, $p(D)=0.14$, $p(A)=0.17$, $p(B)=0.51$

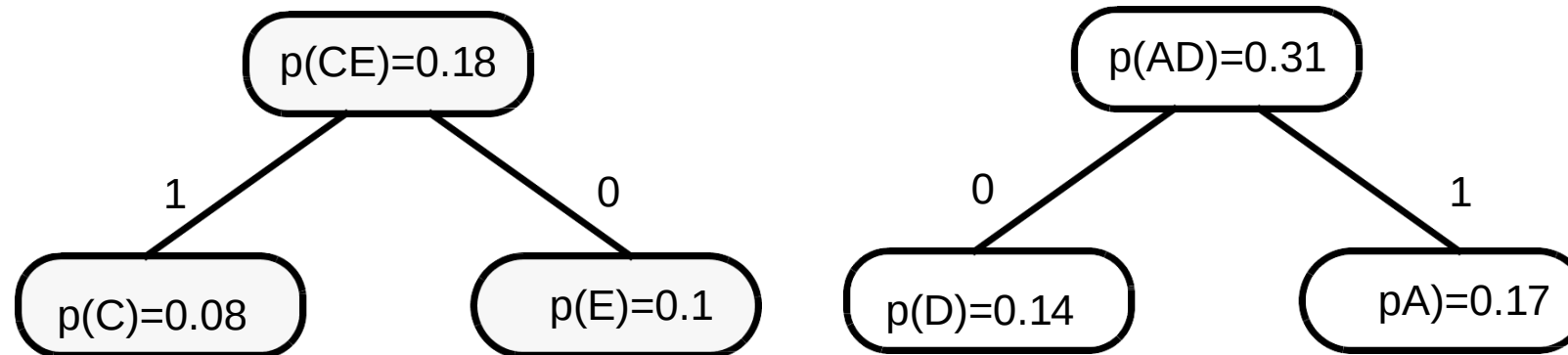


Huffman Encoding—Example

step 2: $p(D)=0.14$, $p(A)=0.17$, $p(CE)=0.18$, $p(B)=0.51$

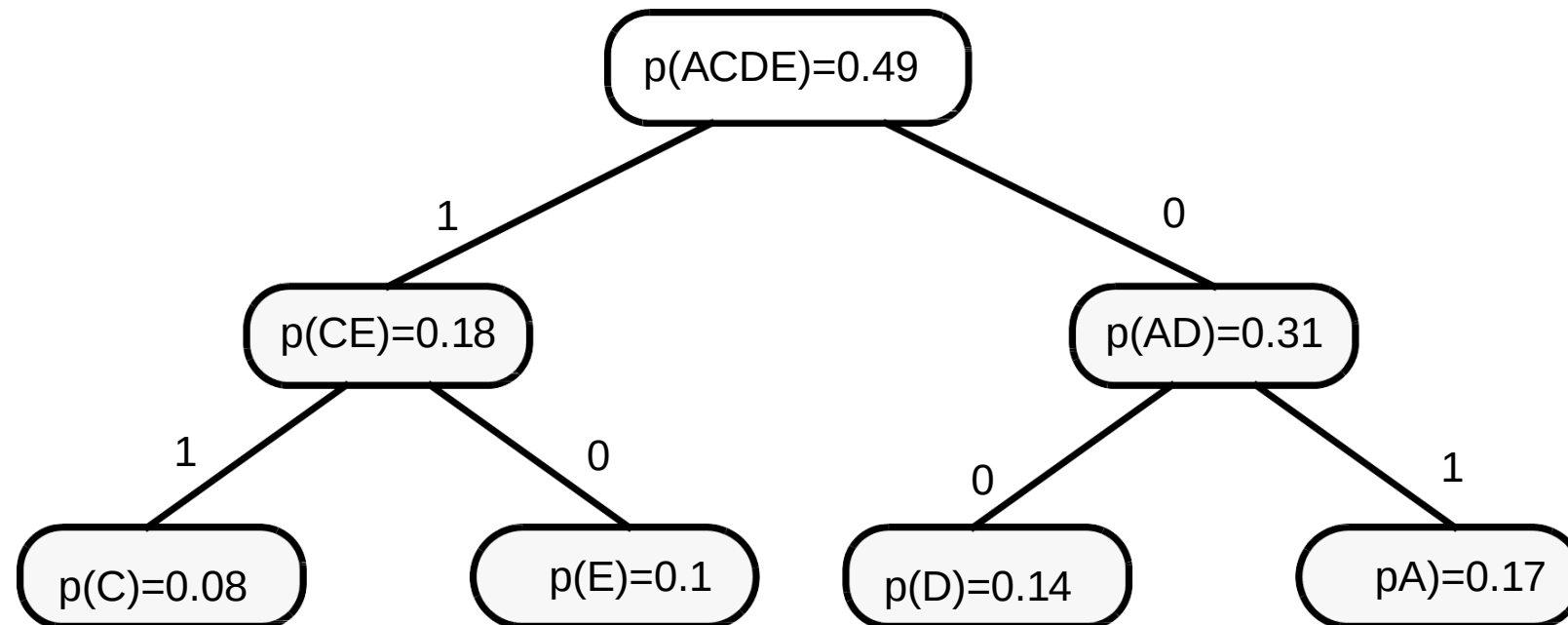
```

build ordered list of symbols (increasing
probability)
do while list contains at least 2 elements
    construct tree using the first two
    elements in list
    add parent node for the union of these
    elements and compute probability
    mark edges by '0' and '1'
    delete the first 2 elements in list;
    insert parent into list
end
    
```



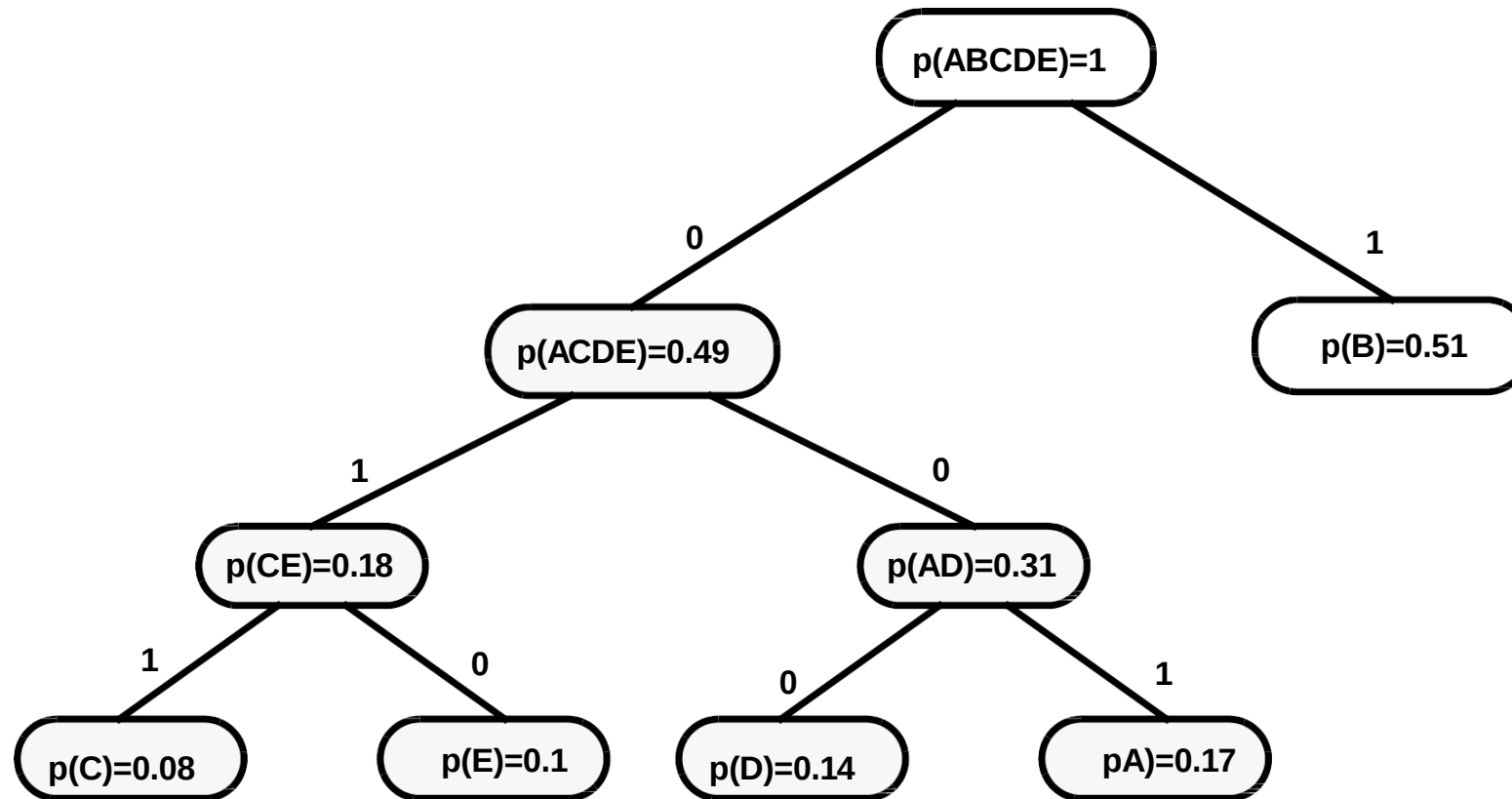
Huffman Encoding—Example

step 3: $p(CE)=0.18$, $p(AD)=0.31$, $p(B)=0.51$



Huffman Encoding—Example

last step: $p(ACDE)=0.49$, $p(B)=0.51$



Huffman Encoding—Example

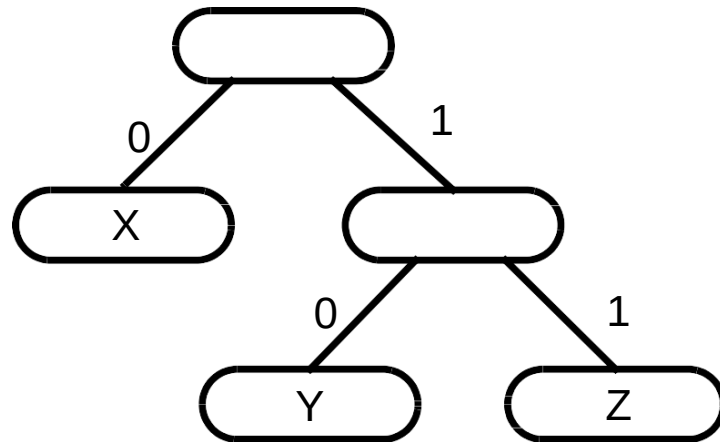
encode symbols according to tree

A	001
B	1
C	011
D	000
E	010

$p(C)=0.08$, $p(E)=0.1$, $p(D)=0.14$, $p(A)=0.17$, $p(B)=0.51$

Huffman Decoding—Example

given: 000101011 and Huffman Tree

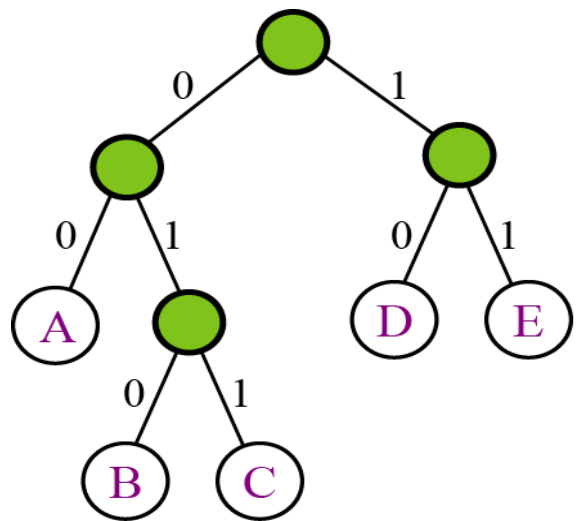


reconstruct message from bit stream using
finite state machine

Solution: XXXYYZ

Huffman Encoding

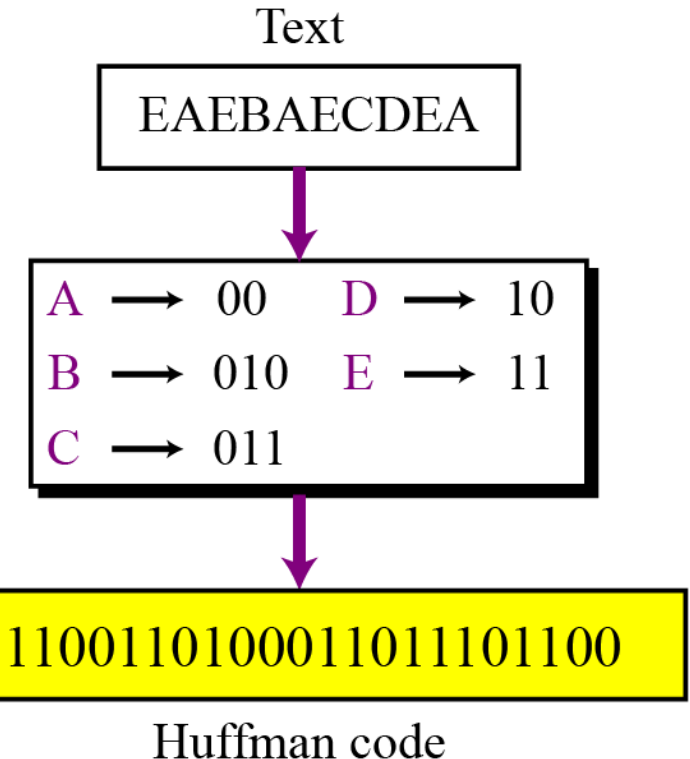
- Character code found by starting at the root and following the branches that lead to that character.
- The code itself is the bit value of each branch on the path, taken in sequence.



A: 00	D: 10
B: 010	E: 11
C: 011	

Code

Encoder



- Decoding: reverse process

MP3 Bandwidth

The MP3 audio format provides lossy data compression. Audio quality improves with increasing bitrate.

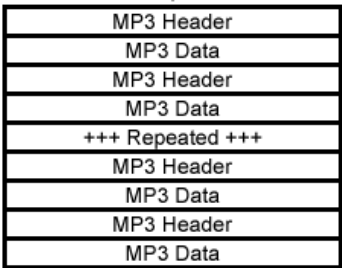
- Constant Bit Rate (CBR) – constant over whole file
- Variable Bit Rate (VBR) – a bit reservoir allows distribution of bit rate over frames
 - lower bit rate for the less complex passages and a higher one for the more complex parts
 - In MP3: Average Bitrate (ABR)
- 32 kbit/s – generally acceptable only for speech
- 96 kbit/s – generally used for speech or low-quality streaming
- 128 or 160 kbit/s – mid-range bitrate quality
- 192 kbit/s – medium quality bitrate
- 256 kbit/s – a commonly used high-quality bitrate
- 320 kbit/s – highest level supported by the MP3 standard



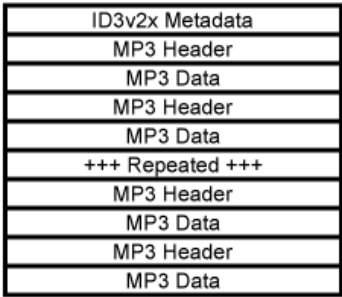
Simple MPEG data stream

Header	CRC	Bit Zuweisung	Skal. Faktor	Sub-band	Werte	Hilfs-daten
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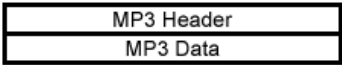
Internal Structure of an MP3 File



Note that the MP3 file structure may be 'encapsulated' within an ID3 tag.



An MP3 Frame



Example MP3 Header

FFFB0400

Colour-coding shows binary bit mapping to hex values below

Detail of an MP3 Header

Bits	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32			
Binary	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0			
Hex	F			F			F						B						A						0				4					0	
																										Mode Extension (Used With Joint Stereo)									
Meaning	MP3 Sync Word												Version	Layer		Error Protection		Bit Rate					Frequency		Pad. Bit	Priv. Bit	Mode				Copy	Original	Emphasis		
																												0 = Intensity Stereo Off	0 = MS Stereo Off	0 = Not Copy- righted	0 = Copy Of Original Media				
Value	Sync Word												1 = MPEG	01 = Layer 3		1 = No		1010 = 160					00 = 44100 Hz		0 = Frame is not padded	Unknown	01 = Joint Stereo						00 = None		

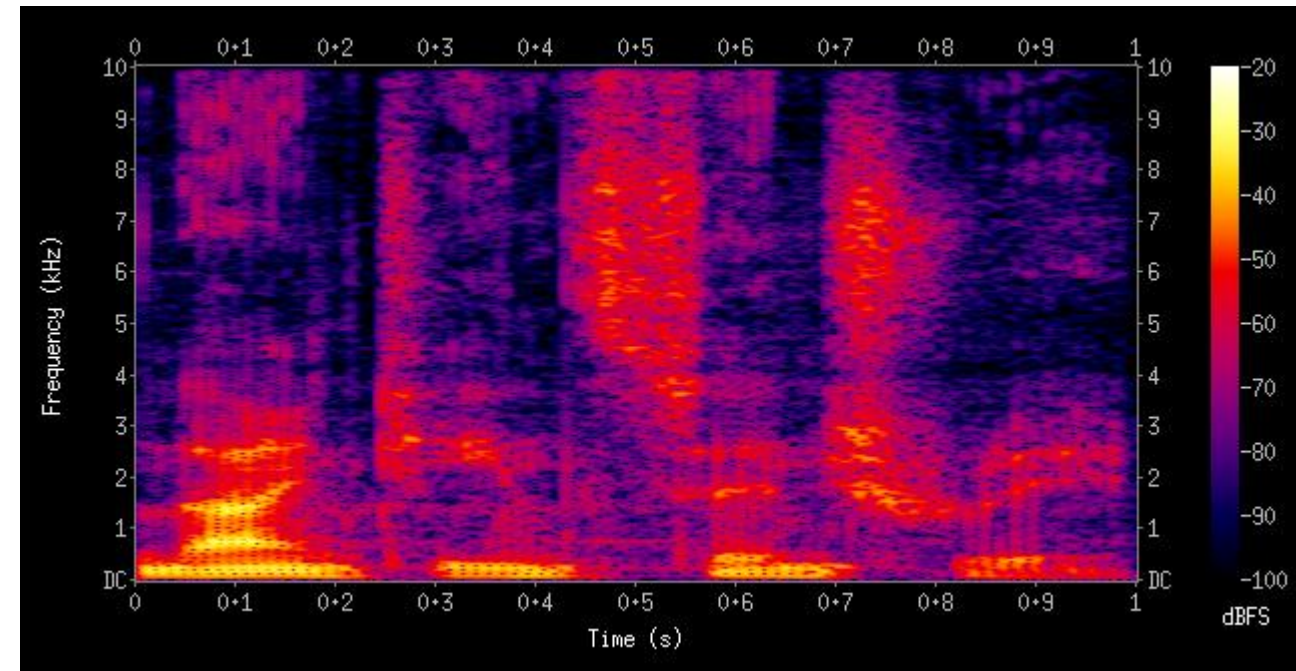
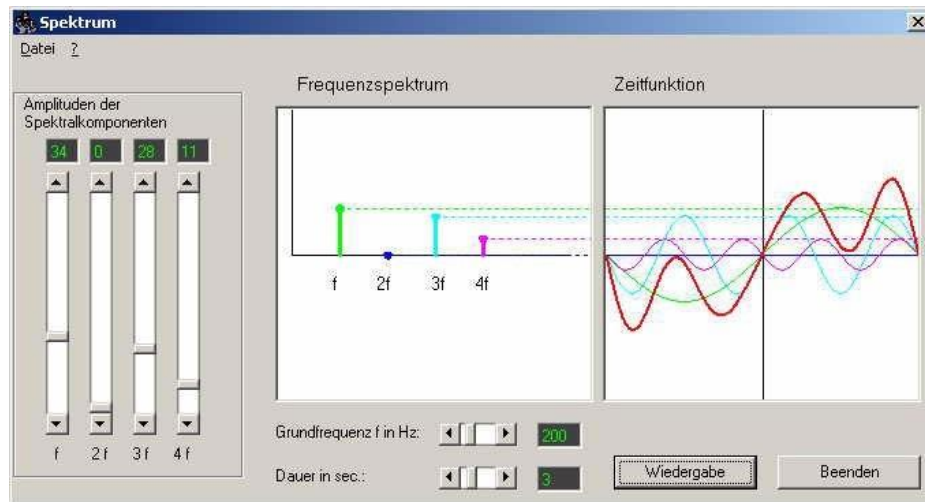
- Header - description of the data stream
- Redundancy Code (CRC) - detection of data stream errors
- Bit assignment - word length of the following subband values
- Scaling factor - 6-bit scaling factor for the following subband values
- Auxiliary data - any data can be inserted

MP3 Further Details

- Further details the algorithm and implementation specifics (based on LAME MP3 encoder)
- <https://wiki.hydrogenaud.io/index.php?title=MP3>

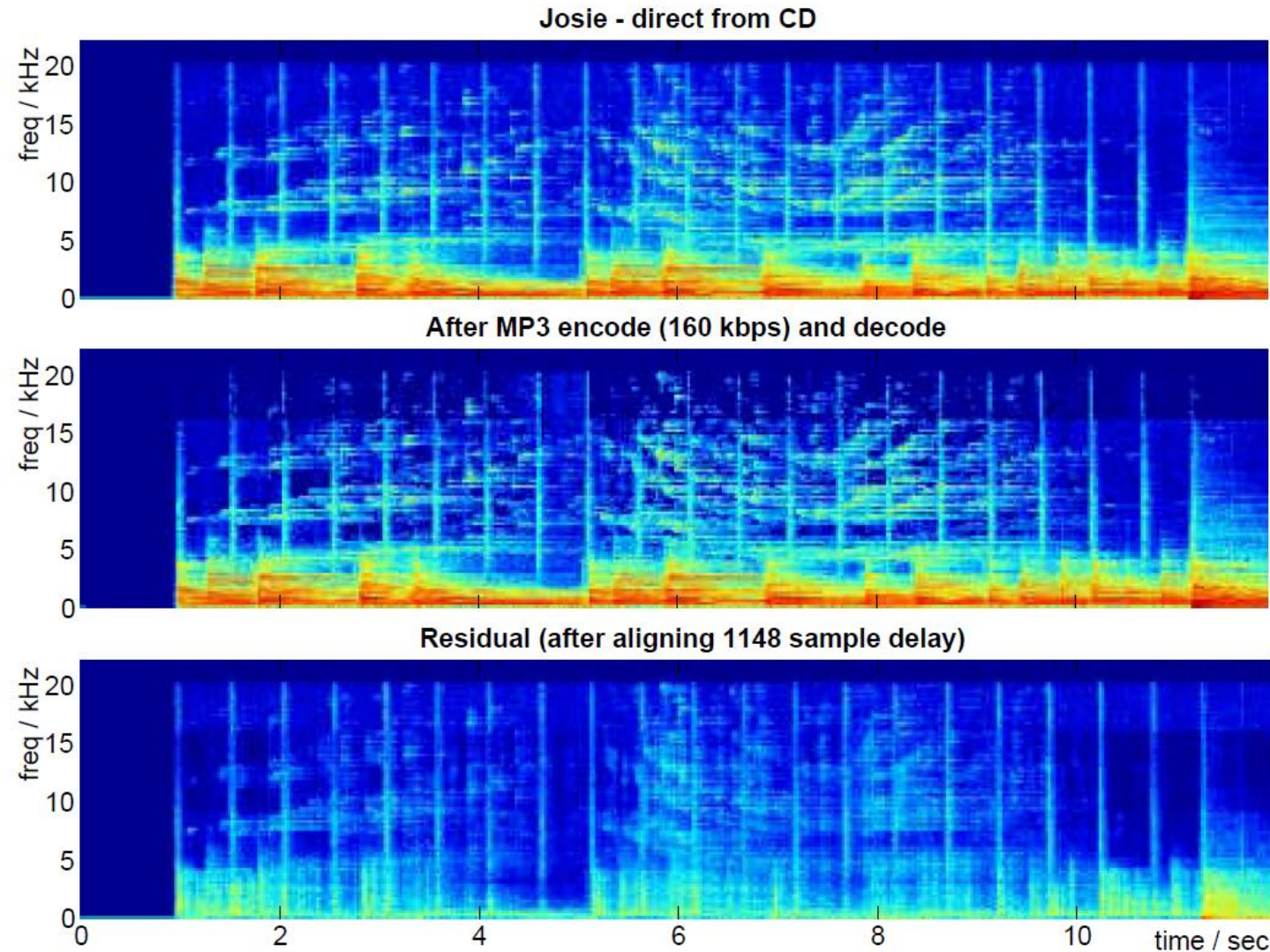
Spectrogram

- A **spectrogram** is a visual representation of the spectrum of frequencies of a signal as it varies with time.



The Effects of MP3

- Chop off high frequency (above 16 kHz)
- Occasional other time-frequency 'holes'
- Quantization noise under signal



Quality Considerations

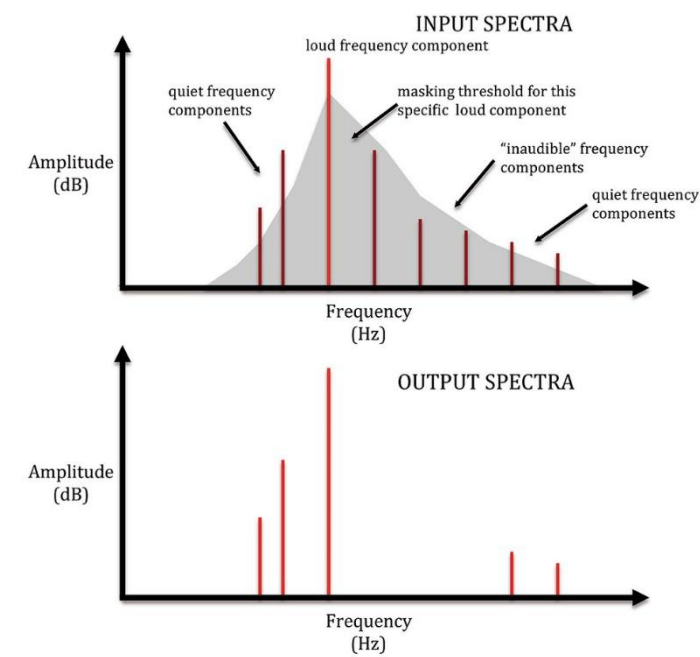
- Common types of artifacts

- loss of bandwidth (no signal $>16\text{kHz}$)
- pre-echoes

Name is misleading: the basic coding artifact is noise spread out over some time even before the music event causing the noise

- roughness, double-speak

at low bit-rates and lower sampling frequencies there is a mismatch between time resolution of the coder and the requirements to follow the time structure of some signals. (e.g. robot like “double-speak”)



Further examples:

<https://www.soundonsound.com/techniques/what-data-compression-does-your-music>

How to Measure Codec Quality

- Simple objective measurement techniques
 - Are **misleading**; these methods defy the whole purpose of perceptual coding
 - Examples of such measurements: signal-to-noise-ratio, bandwidth of decoded signal

How to Measure Codec Quality

Listening tests

- large scale and well-controlled listening tests are still the only method available
- the aim of these tests is to stress the encoders under worst case conditions
- During the ISO-MPEG process: a number of international listening tests with trained listeners.
- tests used the "triple stimulus, hidden reference" (ABC/HR) method and the "CCIR impairment scale"

ABC/HR Test

- listening sequence is "ABC" , with A = original, BC = pair of original / coded signal with random sequence,
- listener must first distinguish the encoded version from the original (Hidden Reference, "HR")
- than assign a score as a subjective judgment of the quality.

CCIR Impairment Scale

- the listener has to evaluate both B and C with a number between 1.0 and 5.0.
- meaning of these values:
 - 5.0 = transparent (this should be the original signal)
 - 4.0 = perceptible, but not annoying (first differences noticable)
 - 3.0 = slightly annoying
 - 2.0 = annoying
 - 1.0 = very annoying

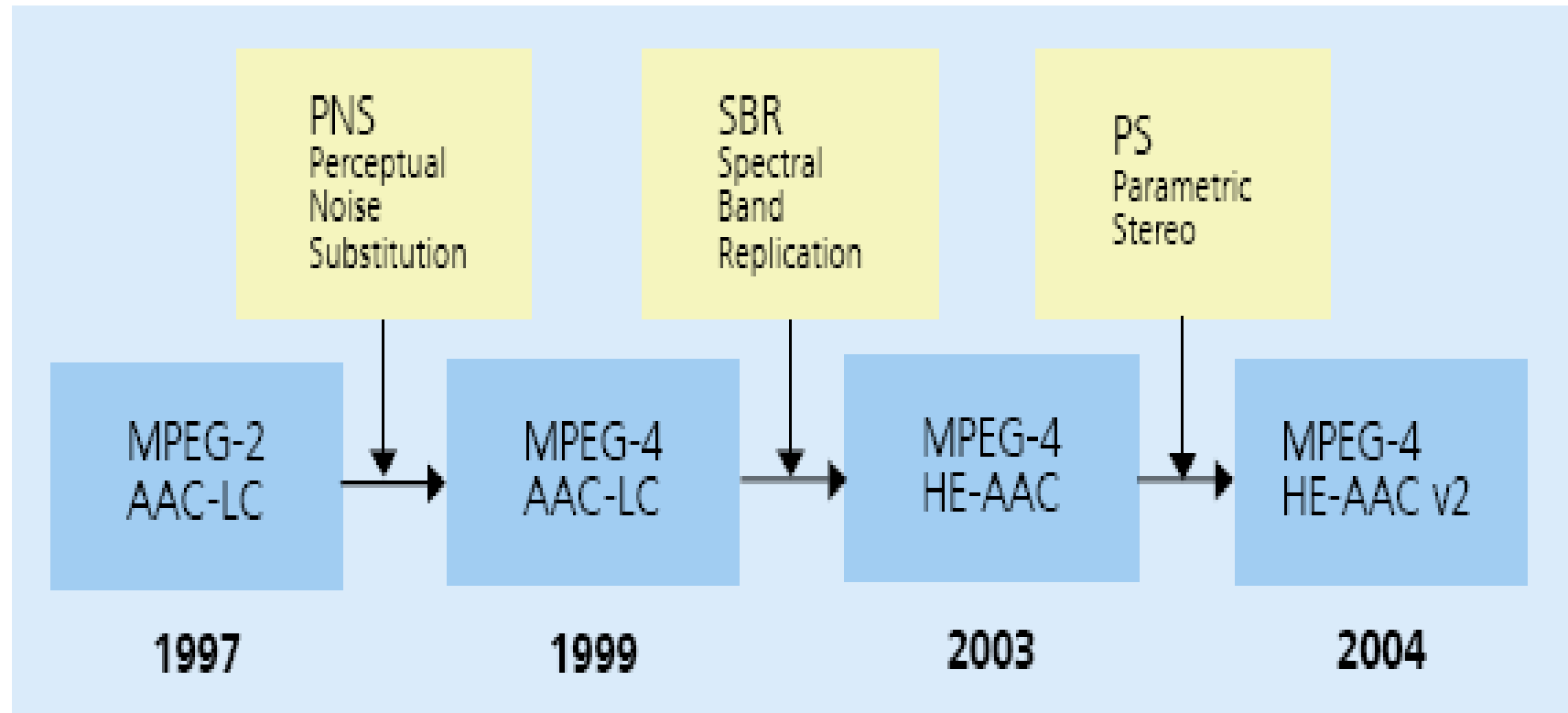
How to Measure Codec Quality

- Perceptual Measurement Techniques (PMT)
 - state of the art not yet sufficient to make large scale and well-prepared listening tests obsolete
 - however, PMT have progressed to the point where they are a very useful supplement to listening tests and can replace them in some cases.
 - ITU-R Task Group 10/4 (Int. Telecommunications Union, Radiocommunications sector) produced a recommendation on a system called PEAQ (Perceptual Evaluation of Audio Quality)
 - a standardized algorithm for objectively measuring perceived audio quality
 - utilizes software to simulate perceptual properties of the human ear and then integrates multiple model output variables into a single metric. PEAQ characterizes the perceived audio quality as subjects would do in a listening test according to ITU-R BS.1116

MPEG-2 Audio

- Original MPEG-2 Audio standard (1995) consists of 2 extensions to MPEG-1:
 - backwards compatible multichannel coding adds option of forward and backwards compatible coding of multichannel signals (including 5.1 channel configurations)
 - Coding at lower sampling frequencies of 16kHz, 22.05 kHz and 24 kHz
- Verification tests in 1994 showed that introducing new coding algorithms and giving up backwards compatibility promised a significant improvement in coding efficiency => AAC

MPEG AAC FAMILY



Advanced Audio Codec (AAC)

- successor of MP3 format
- Specified in MPEG-2 (Part 7) (1997) and MPEG-4 (Part 3) (1999)
- AAC generally achieves higher sound quality than MP3 at the same bit rate
- HE-AAC and HE-AAC v2 (in MPEG-4/Part 3)
 - extensions of the original AAC, but do not replace it
 - are optimized exclusively for low bit rates and are used, for example, for live broadcasts of digital radio and television.
 - At higher bit rates (from 96 kbit/s), they perform worse than AAC and should not be used

AAC's Improvements over MP3 1/2

- More sample rates (from 8 to 96 kHz) than MP3 (16 to 48 kHz);
- Up to **48 channels** (MP3 supports up to two channels in MPEG-1 mode and up to 5.1 channels in MPEG-2 mode);
- Arbitrary bit rates and **variable frame length**. Standardized constant bit rate with bit reservoir;
- Higher efficiency and **simpler filter bank**. AAC uses a pure **MDCT** (modified discrete cosine transform), rather than MP3's hybrid coding (which was part MDCT and part FFT);
- Higher coding efficiency for stationary signals (AAC uses a blocksize of 1024 or 960 samples, allowing more efficient coding than MP3's 576 sample blocks);
- Higher coding accuracy for transient signals (AAC uses a blocksize of 128 or 120 samples, allowing more accurate coding than MP3's 192 sample blocks);
- Possibility to use Kaiser-Bessel derived window function to eliminate spectral leakage at the expense of widening the main lobe;

AAC's Improvements over MP3 2/2

- Much better handling of **audio frequencies above 16 kHz**;
- More flexible joint stereo (different methods for different frequency ranges)
- Additional modules (tools) added to increase compression efficiency:
Temporal Noise Shaping (TNS), backwards prediction, perceptual noise substitution (PNS), etc.
 - Modules can be combined to constitute different encoding profiles
- Overall, AAC format allows developers more flexibility to design codecs than MP3 does, and corrects many of the design choices made in the original MPEG-1 audio specification. This **increased flexibility** often leads to more concurrent encoding strategies and more efficient compression – especially at **very low bit rates** where superior stereo coding, pure MDCT, and better transform window sizes leave MP3 unable to compete.

Advantages Over MP3

AAC

1. Multi Channel Audio – up to 48 audio channels
2. Sample frequencies from 8KHz - 96KHz
3. Simpler filter bank (pure MDCT used)
4. Better stationary and transient response due to block sizes of 1024 and 128 samples
5. Excellent handling of high frequency signals
6. CD quality audio at 64Kbits/sec
7. Much better quality of audio at lower bit rates (down to 32Kbps)

MP3

1. Stereo signal – maximum of only 2 channels
2. Sampling frequencies from 16KHz ~ 48KHz
3. Hybrid filter bank (requires more computational power)
4. Poorer stationary and transient response due to block sizes of 576 and 192 samples
5. Signal handling up to 15.5/15.8 KHz
6. CD quality audio at 128Kbits/sec
7. Audio quality is poorer at low bit rates and may present coding artifacts

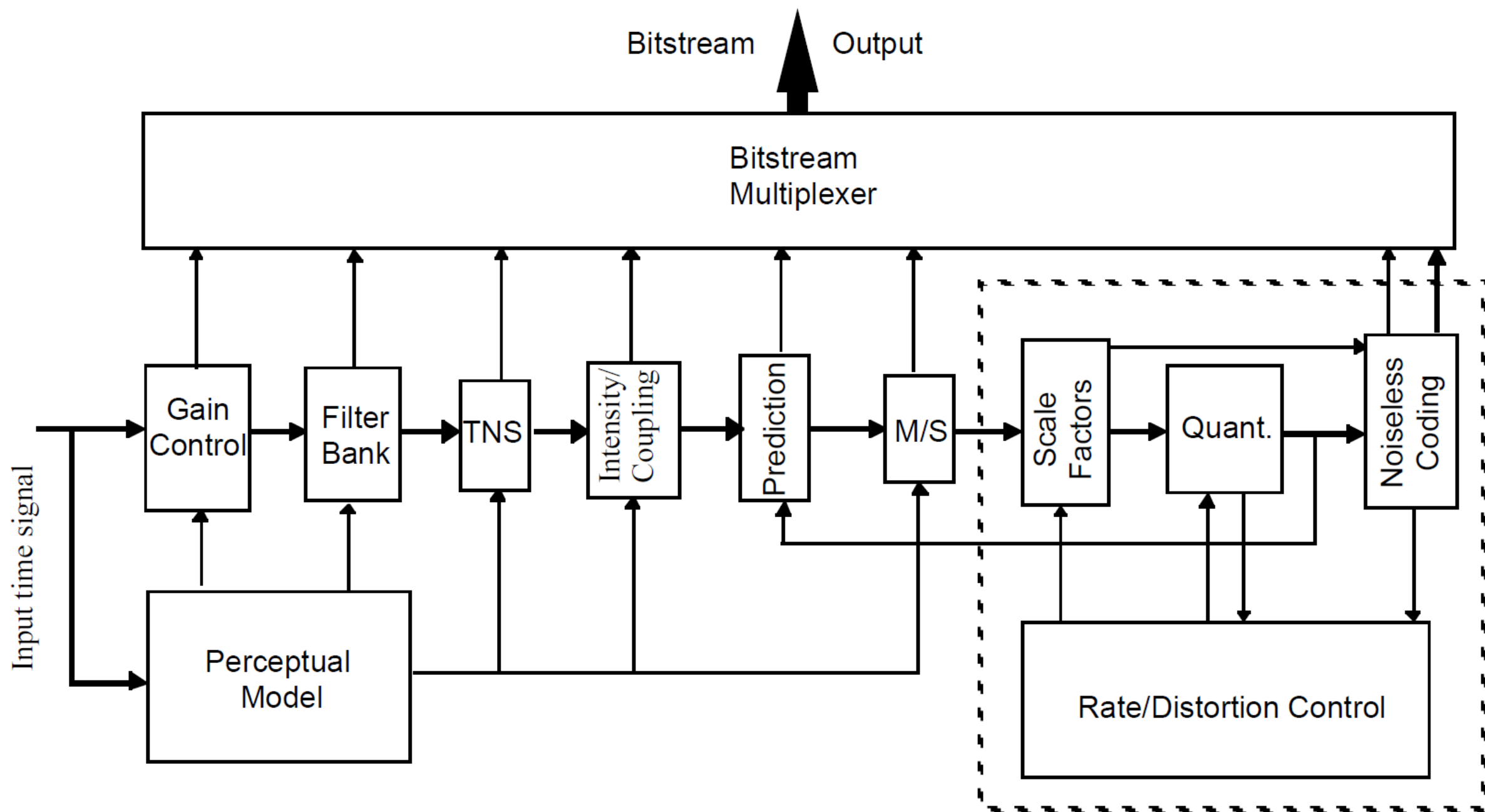


Figure 3: Block diagram of an MPEG-2 AAC encoder.

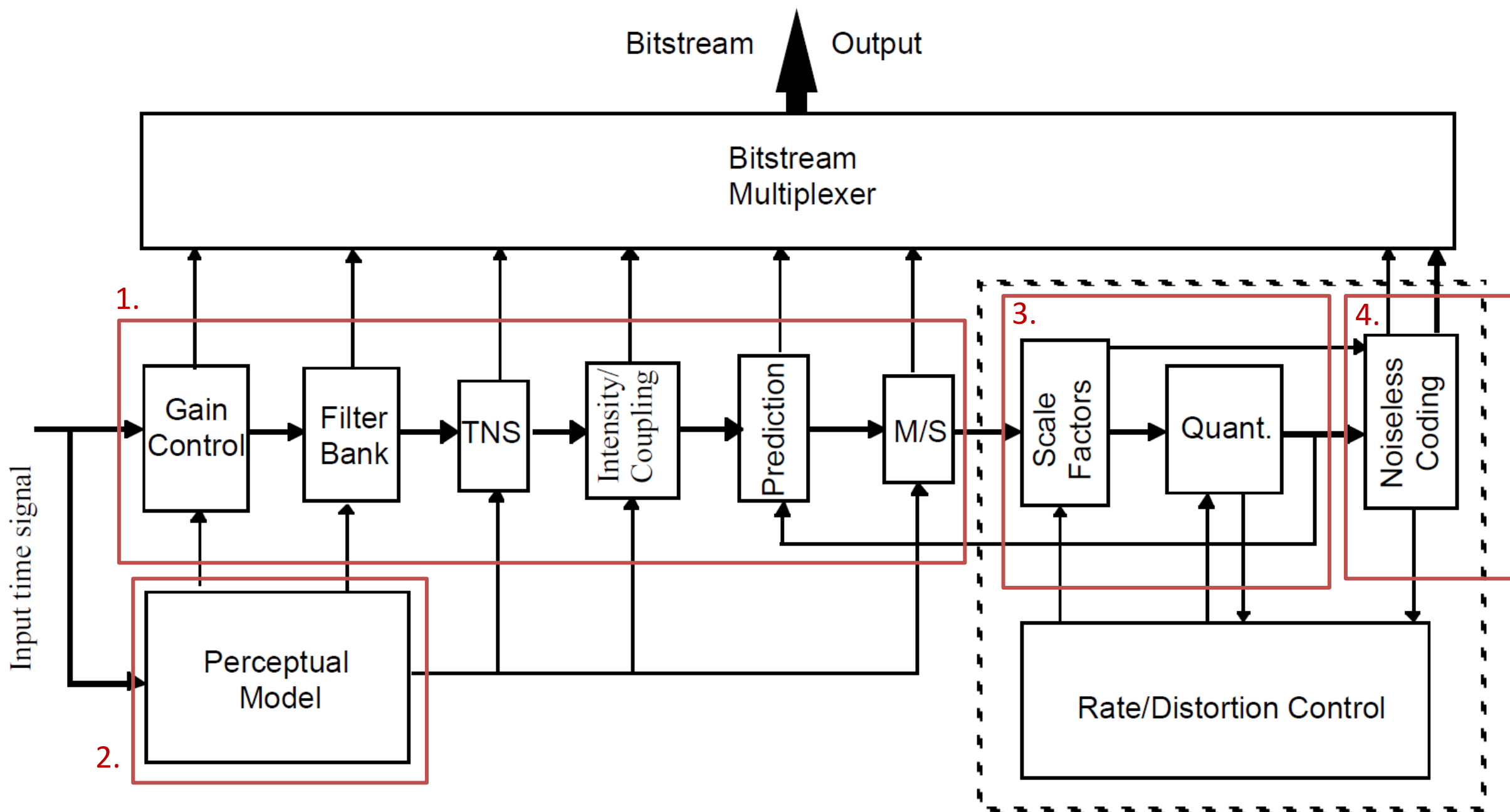
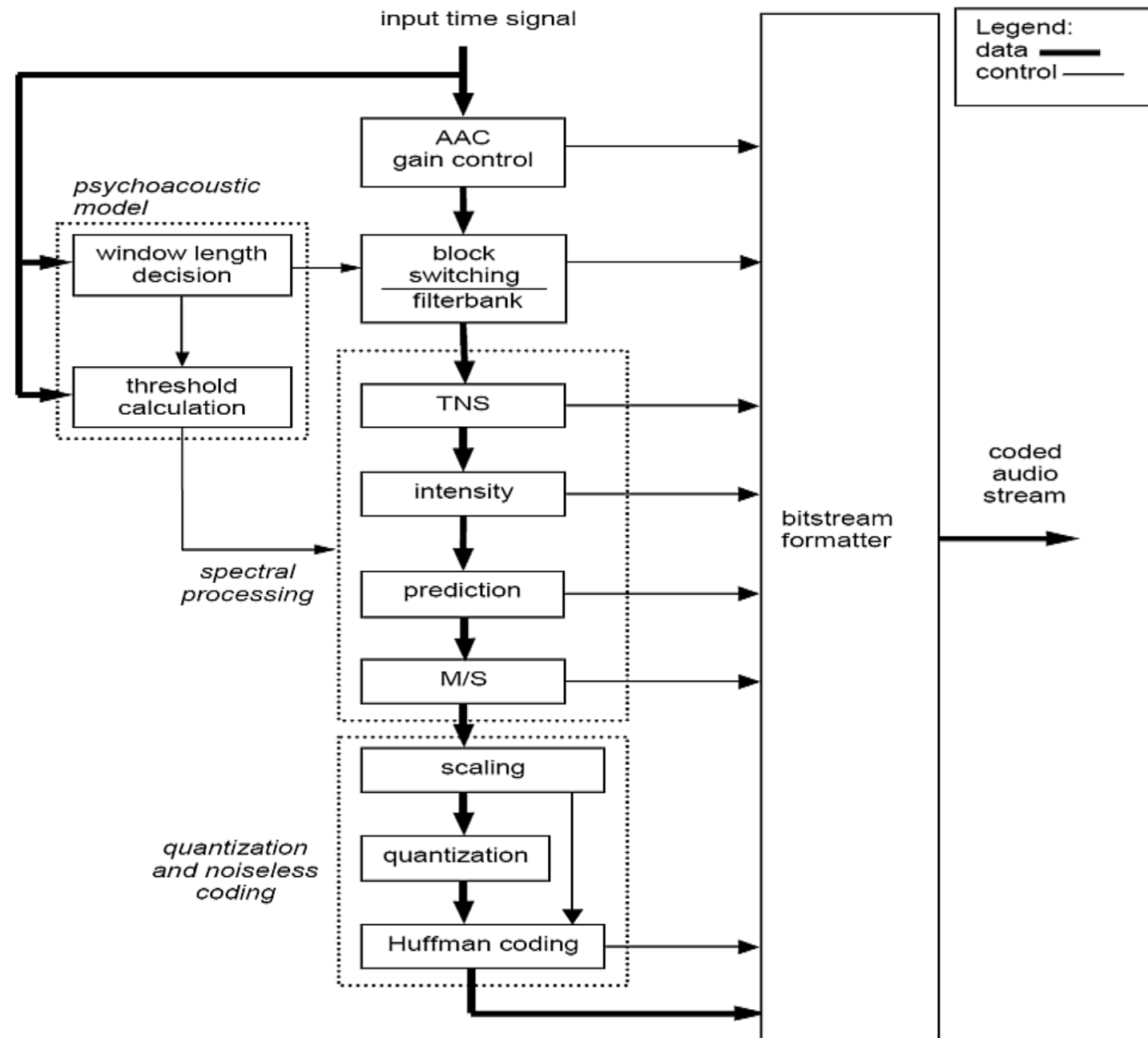
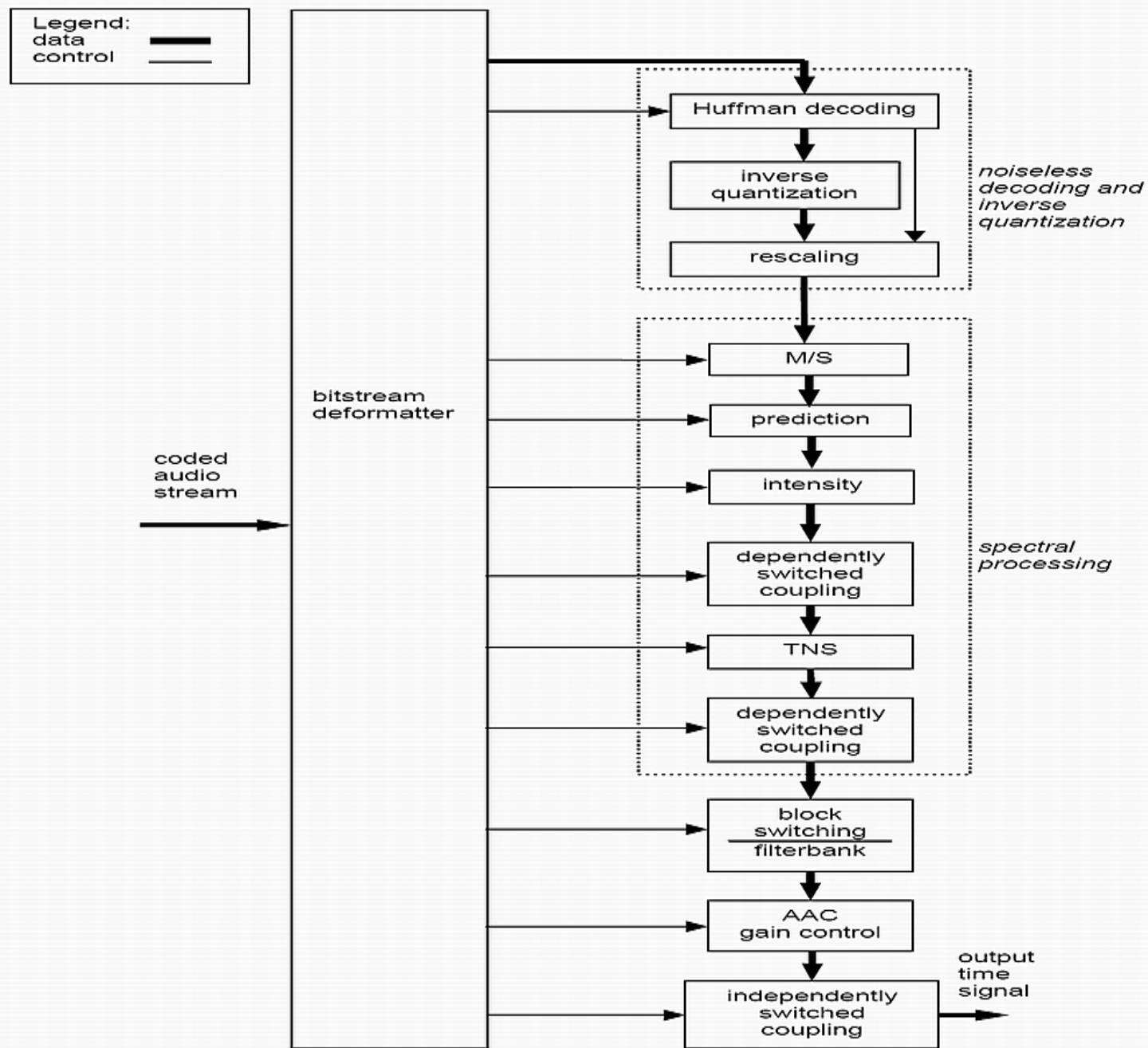


Figure 3: Block diagram of an MPEG-2 AAC encoder.



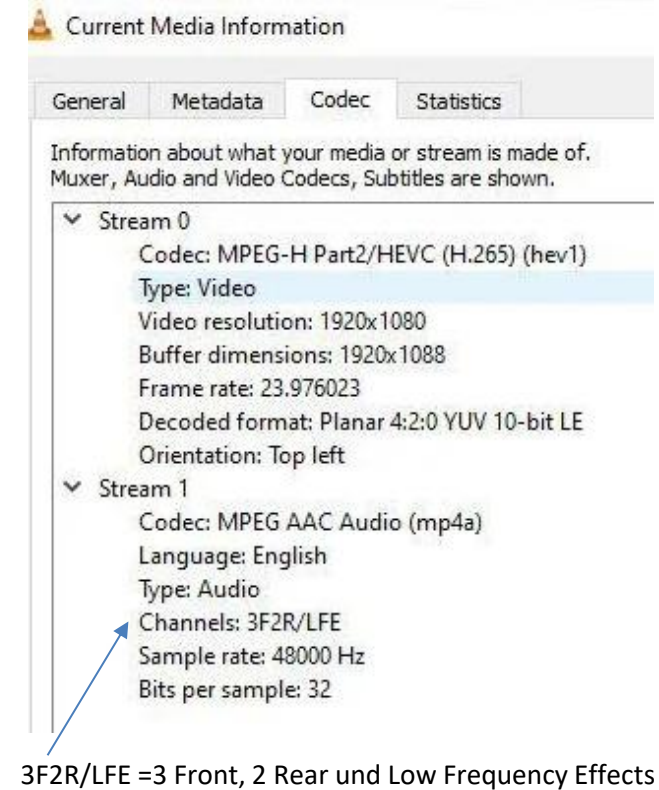
MPEG-2 AAC Encoder Block Diagram



MPEG-2 AAC Decoder Block Diagram

Dolby Digital 1/2

- Dolby Digital contains up to six discrete channels of sound. 5.1 mode has
 - five channels for normal-range speakers (20 Hz – 20,000 Hz) (right, center, left, right surround, left surround) and
 - one channel (20 Hz – 120 Hz allotted audio) for the subwoofer driven low-frequency effects.
 - Mono and stereo modes are also supported.
- Dolby **AC-3** (1991) multi-channel audio coding standard = *ATSC A/52* or *AC-3 Bitstream-Format (Adaptive Transform Coder 3)*
 - Basis is the modified discrete cosine transform (MDCT), a lossy audio compression algorithm. It is a modification of the discrete cosine transform (DCT) algorithm.
 - AC-3 supports audio sample-rates up to 48 kHz.
- Patents expired (since 2017), free to use

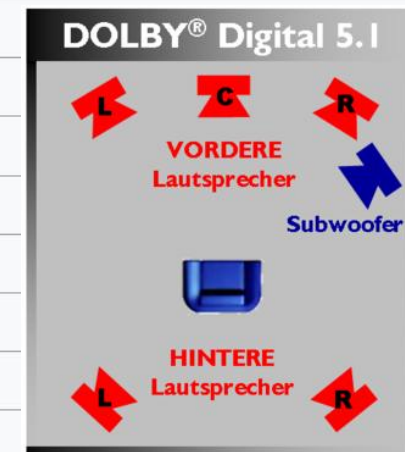


3F2R/LFE = 3 Front, 2 Rear und Low Frequency Effects

Dolby Digital 2/2

Mögliche Kanalkonfigurationen bei Dolby Digital

Name	Modus	Vorne			Hinten		Subwoofer (LFE)	Verwendung
		Links	Mitte (Center)	Rechts	Surround Links	Surround Rechts		
Dolby Digital 1/0	Mono	Nein	Ja	Nein	Nein	Nein	Nein	Vor allem alte restaurierte Filme.
Dolby Digital 2/0	Stereo	Ja	Nein	Ja	Nein	Nein	Nein	Audiokommentar. Kann auch Dolby Surround Informationen enthalten.
Dolby Digital 3/0	3-Stereo	Ja	Ja	Ja	Nein	Nein	Nein	
Dolby Digital 2/1	Stereo mit Mono-Surround	Ja	Nein	Ja	Ja		Nein	
Dolby Digital 2/1.1	Stereo mit Mono-Surround und LFE	Ja	Nein	Ja	Ja		Ja	
Dolby Digital 3/1	3-Stereo mit Mono-Surround	Ja	Ja	Ja	Ja		Nein	
Dolby Digital 3/1.1	3-Stereo mit Mono-Surround und LFE	Ja	Ja	Ja	Ja		Ja	
Dolby Digital 2/2	Quadrophonie	Ja	Nein	Ja	Ja	Ja	Nein	
Dolby Digital 2/2.1	Quadrophonie mit LFE	Ja	Nein	Ja	Ja	Ja	Ja	
Dolby Digital 3/2	5-Kanal-Surround	Ja	Ja	Ja	Ja	Ja	Nein	
Dolby Digital 3/2.1	5-Kanal-Surround mit LFE	Ja	Ja	Ja	Ja	Ja	Ja	Meistens als Haupttonspur bei Spielfilmen oder auch Serien eingesetzt.



Extensions to 7.1 (2003) and 9.1 (2006) speaker configurations, resulting in Dolby Surround.

Spatial Audio

Binaural Audio

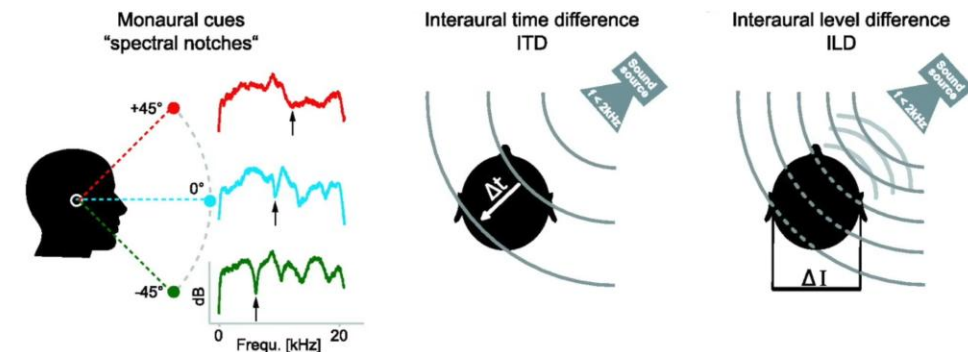
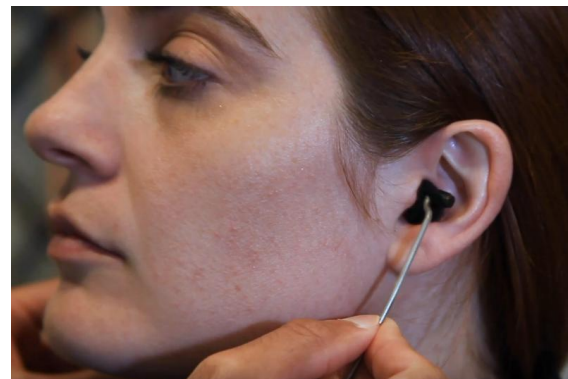
- Spatialize (=render) audio in 3D for headphones
 - more advanced processing than stereo
 - Sends individual signals to left/right ear
 - Uses psychoacoustics to spatialize sound



<https://www.youtube.com/watch?v=FMdabRXWwDE&list=PL7w4cOVVxL6FTg6i9osmlWK-w2wU4xvI2&index=5>

Head Related Transfer Function (HRTF)

- characterizes how an ear receives a sound from a point in space
- Different for every individual



Example: Virtual Barber Shop
<https://www.youtube.com/watch?v=IUDTlvagjJA>

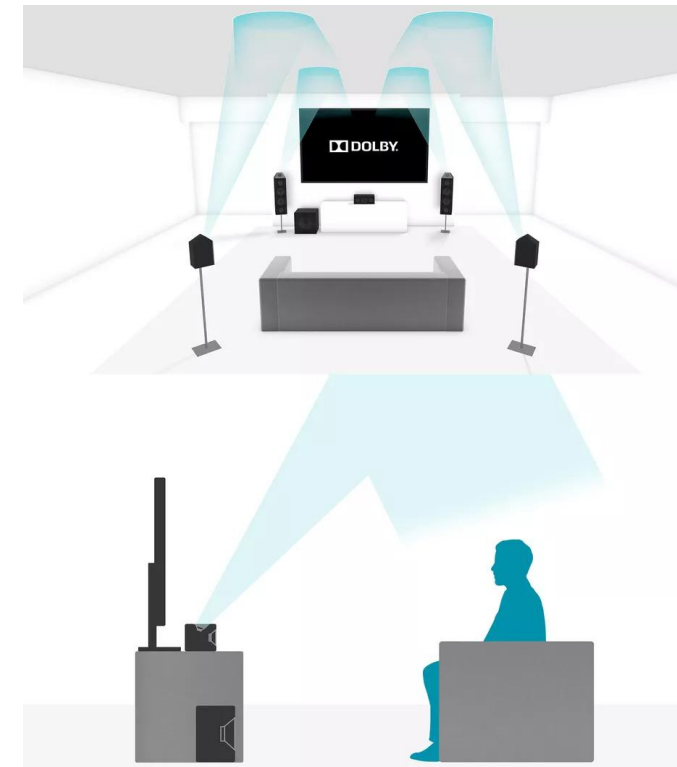
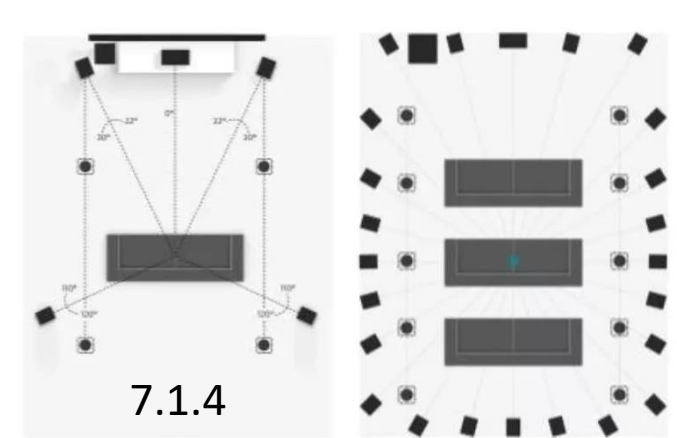
<https://www.youtube.com/watch?v=cB-IW9KKu4>

<http://physrev.physiology.org/content/90/3/983.long>



Dolby Atmos

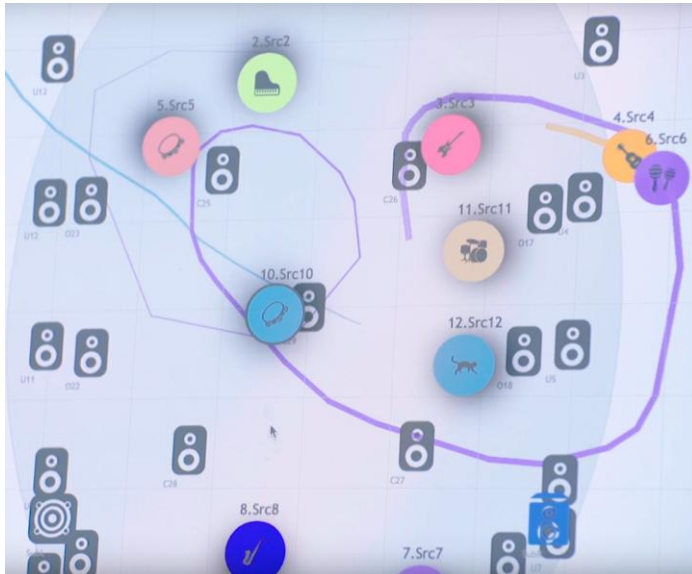
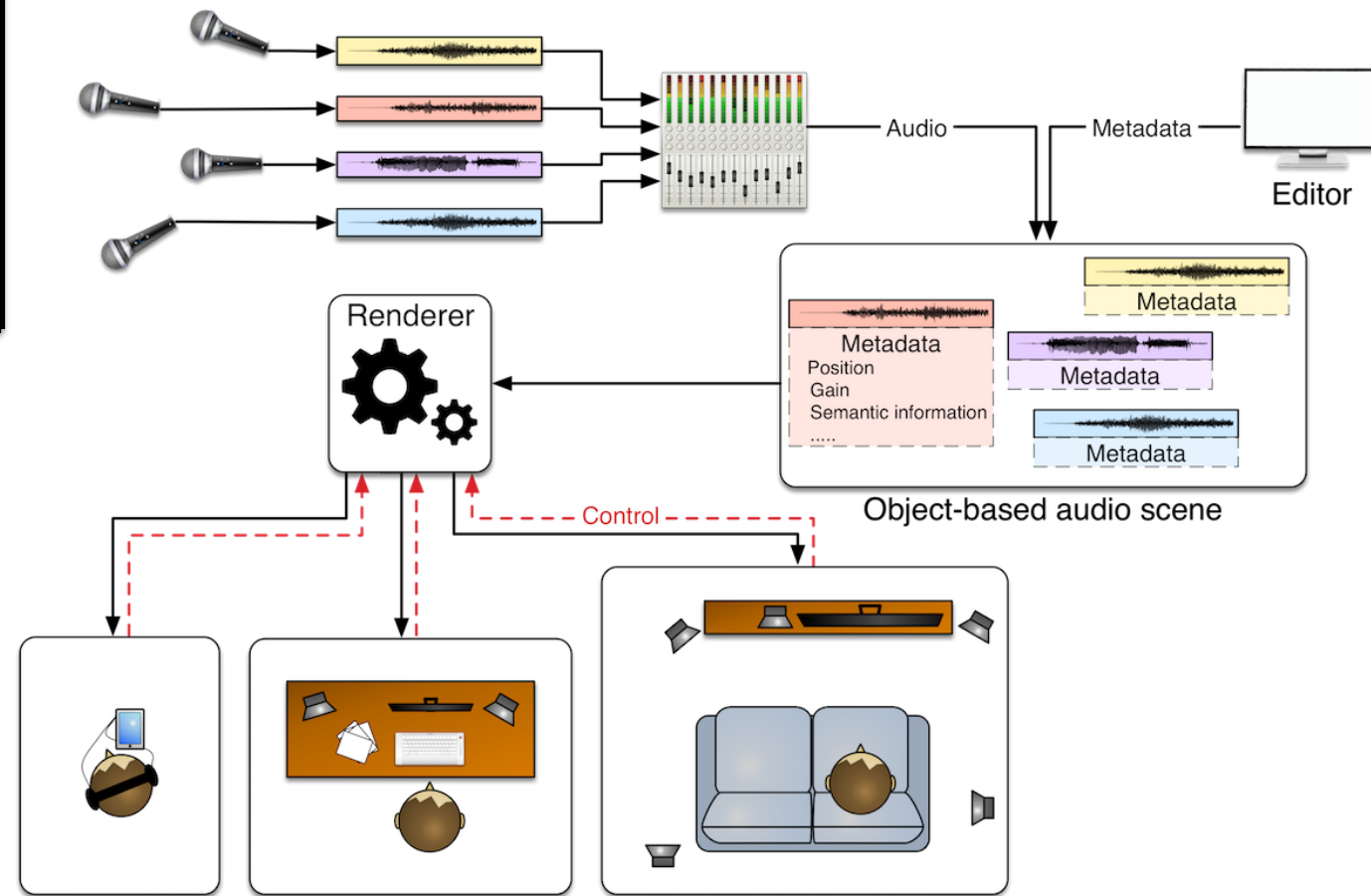
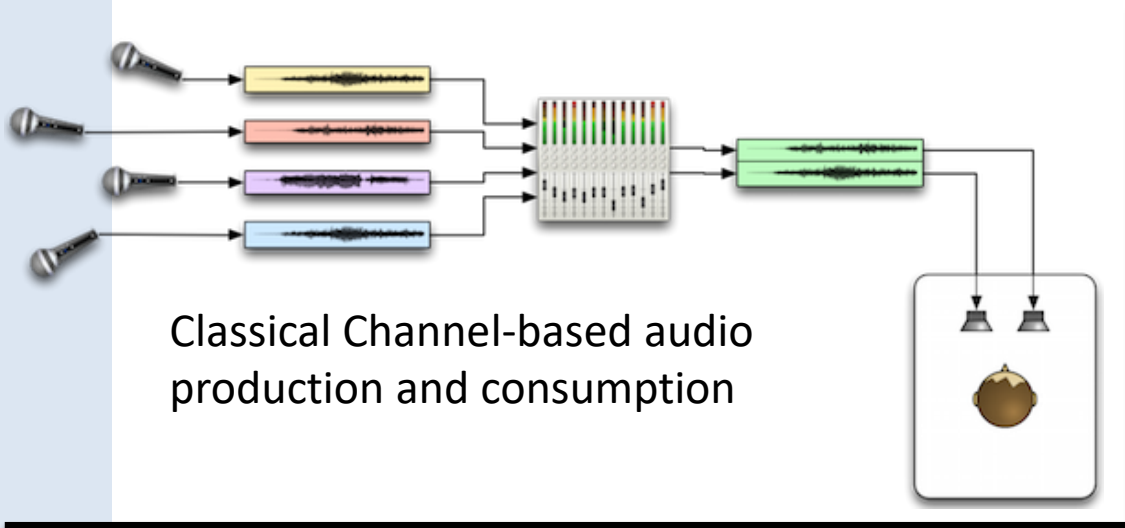
- Up to 128 **audio objects** (or channels)
- For home theaters typically 5.1 or 7.1 base channels with overhead or “Dolby Atmos speakers” (projecting to ceiling) e.g. 7.1.4 (7.1 + 4 overhead)
- Audio objects allow specification of location of sound – rendered in real time based on speaker positions
- Dolby-TrueHD-Codec – **lossless** – based on
 - Meridian Lossless Packing, also known as Packed PCM (PPCM) or Dolby Lossless for compressing PCM audio data
 - 1.5-3 : 1 compression rates (uses redundancy in multiple channels); incl. error correction up to 2ms
 - 44.1-192 kHz, up to 24 bits, 64 channels



Audio Objects



<https://www.youtube.com/watch?v=Y79Q1sfvNmc>

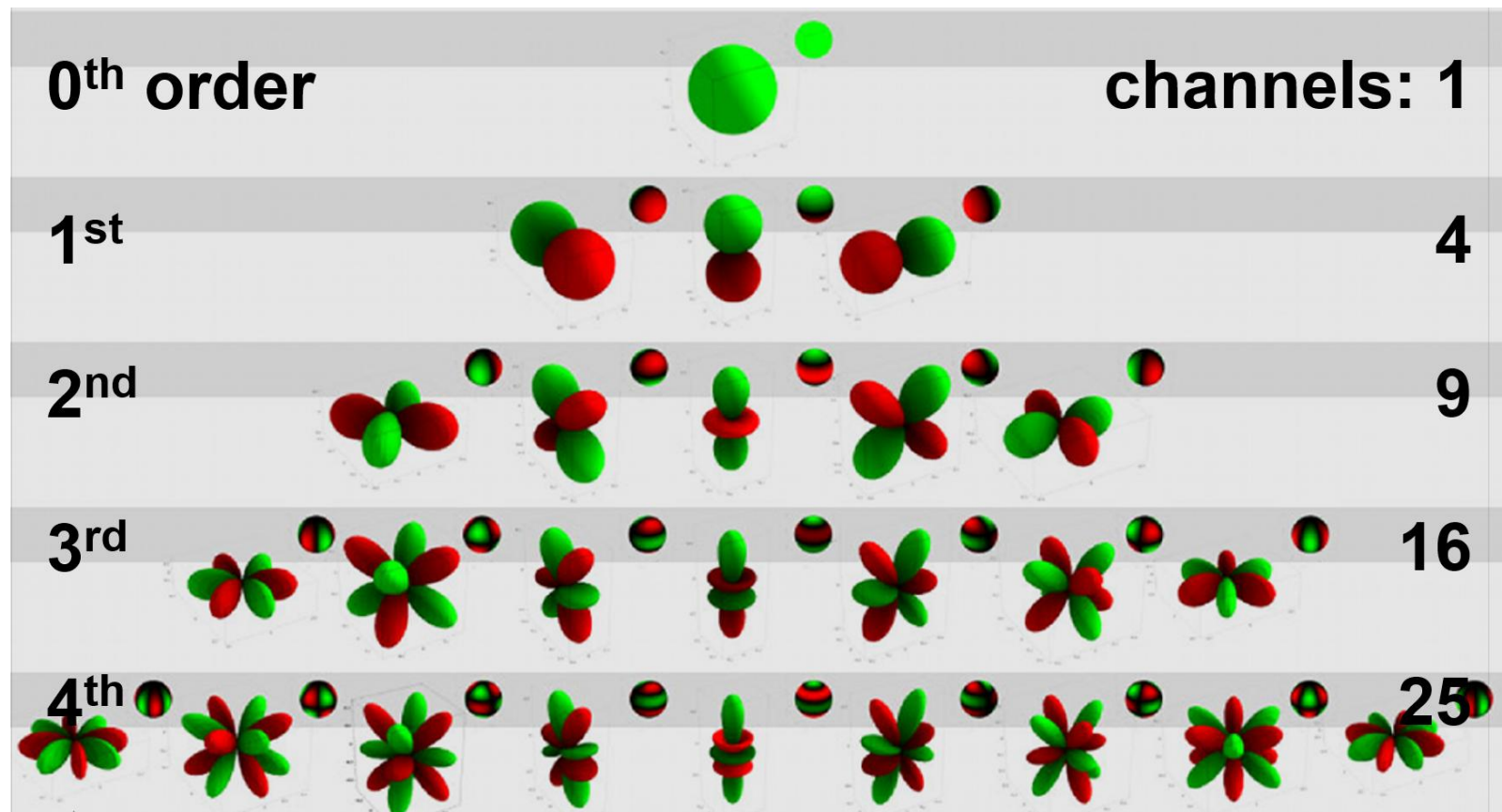


<https://www.youtube.com/watch?v=cqC1rWSxT4c>

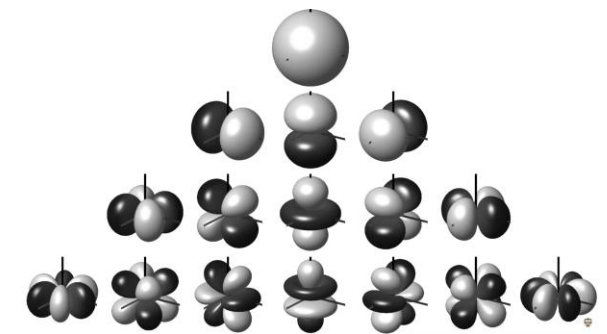


Spatial Audio: Higher-Order Ambisonics

- Full-sphere surround sound format



3rd order Ambisonics microphone
19 microphone omnidirectional
digital MEMS capsules on a sphere



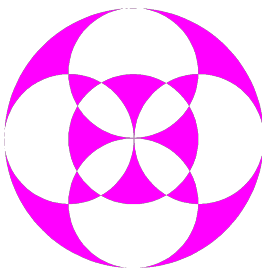
<https://www.youtube.com/watch?v=9m4VxFSYSEA>

Mono, Stereo, 3rd order Ambisonics compared

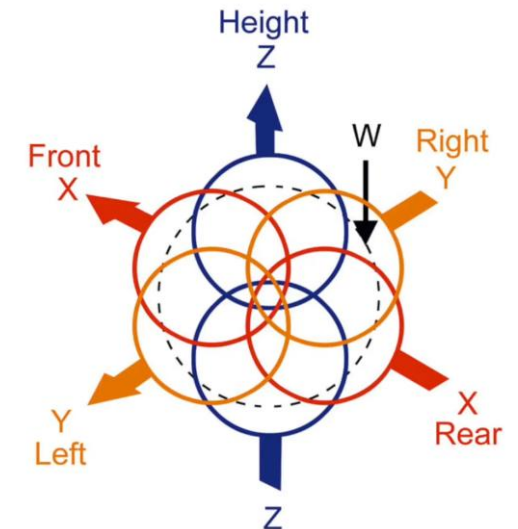
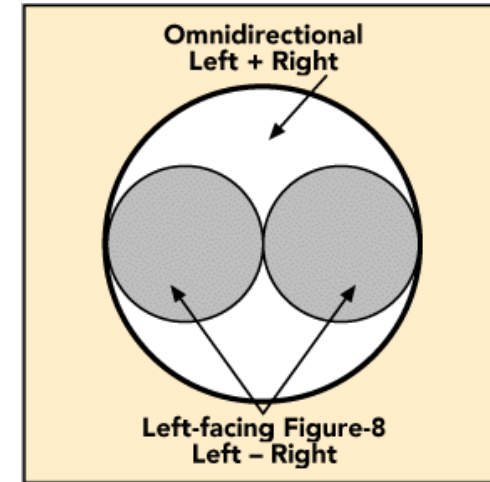


https://www.youtube.com/watch?v=an9N4CKCO_o

Principles of Ambisonics



- Taking mid-side (M-S) coincident pair scheme into three dimensions
- **A-Format** Recording: All individual microphone channels
- **B-Format**: Processed individual channels e.g.
- 1st order Ambisonics **B-Format**:
 - $W = \text{Omni (F+B+L+R+U+D)}$
 - $X = \text{Front} - \text{Back}$
 - $Y = \text{Left} - \text{Right}$
 - $Z = \text{Up} - \text{Down}$
- These signals are symbolized in the Ambisonics logo
- 2nd order Ambisonics uses 9 channels
- 3rd order Ambisonics jumps up to 16 channels
- ... 6th order Ambisonics with 49 channels



What Is Ambisonics?

- “Ambisonics” means “Surround Sound”
- Surround system developed by a team of British researchers in the 1970s, notably the late Michael Gerzon. All patents expired.
- Designed to bring the concert hall into the living room, by accurately capturing and replaying a soundfield on practical speaker arrays
- Four channels (B-Format) (in case of 1st order Ambisonics) capture entire soundfield – including height (“Periphony”)
- Transmission systems compatible with stereo (UJJ) and modern “discrete” surround (G-Format)
- Ambisonic production can use special microphone (Soundfield mic) or multitrack sources

The Kirchhoff-Helmholtz Integral

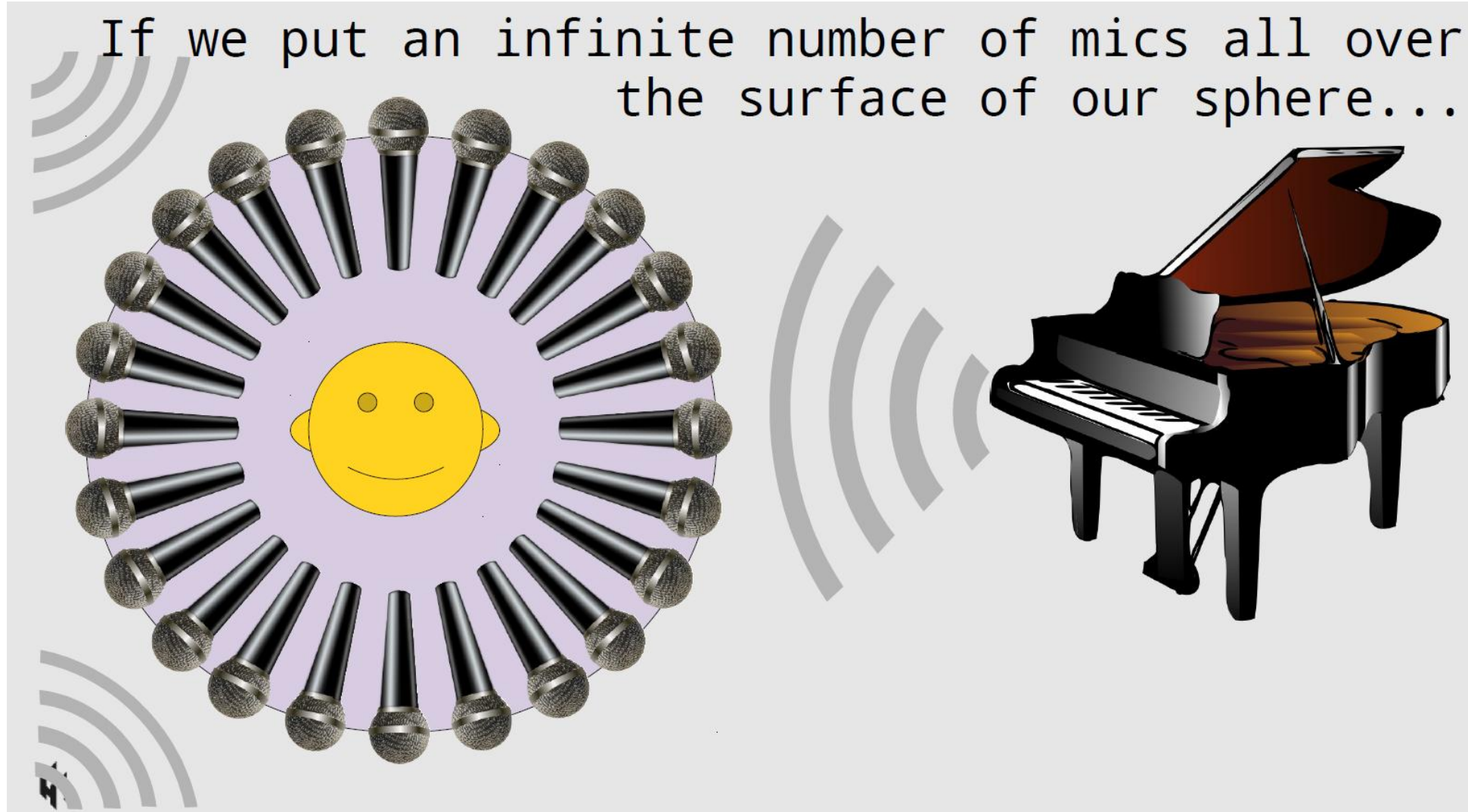
= If you can record the boundary,

"If you know the sound pressure and velocity in any point on the surface of a volume free of sources, you have complete knowledge of the sound field inside."

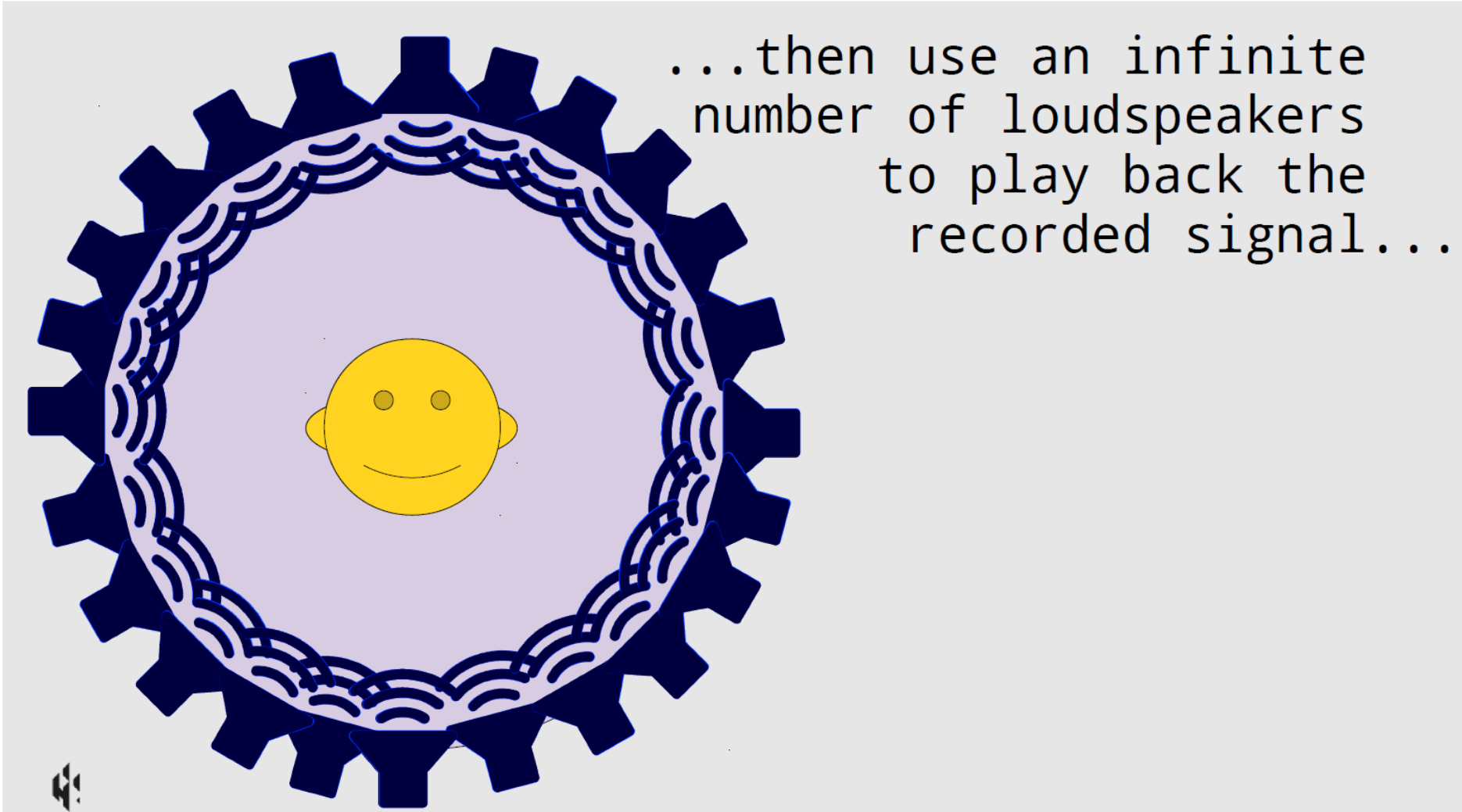
you can reproduce the inside.

The Kirchhoff-Helmholtz Integral

If we put an infinite number of mics all over the surface of our sphere...

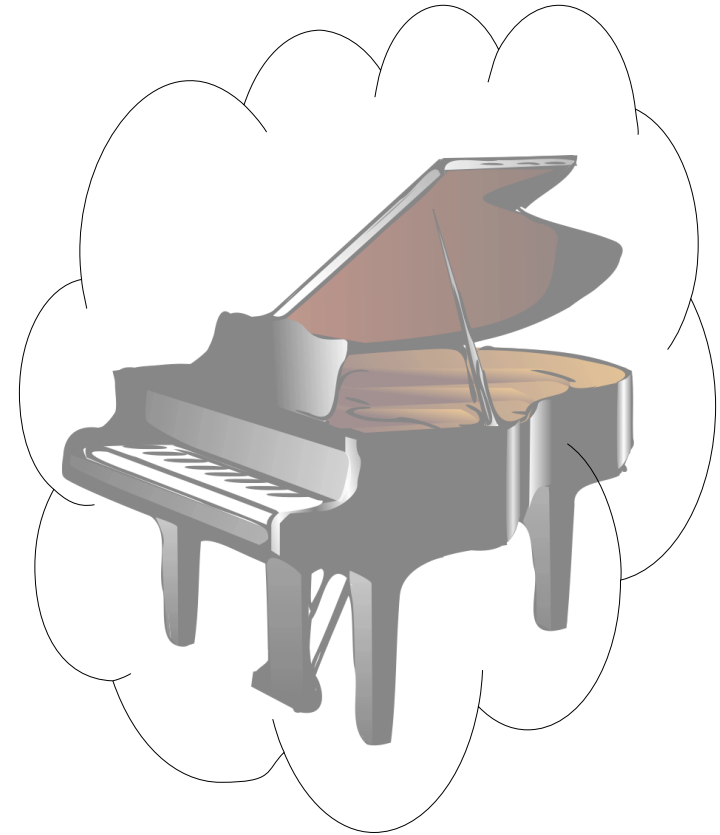
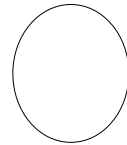
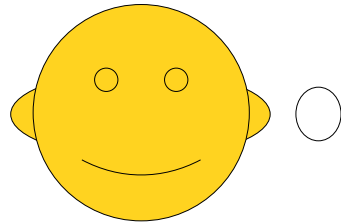


The Kirchhoff-Helmholtz Integral



The Kirchhoff-Helmholtz Integral

...the result will be
identical to the
original performance.



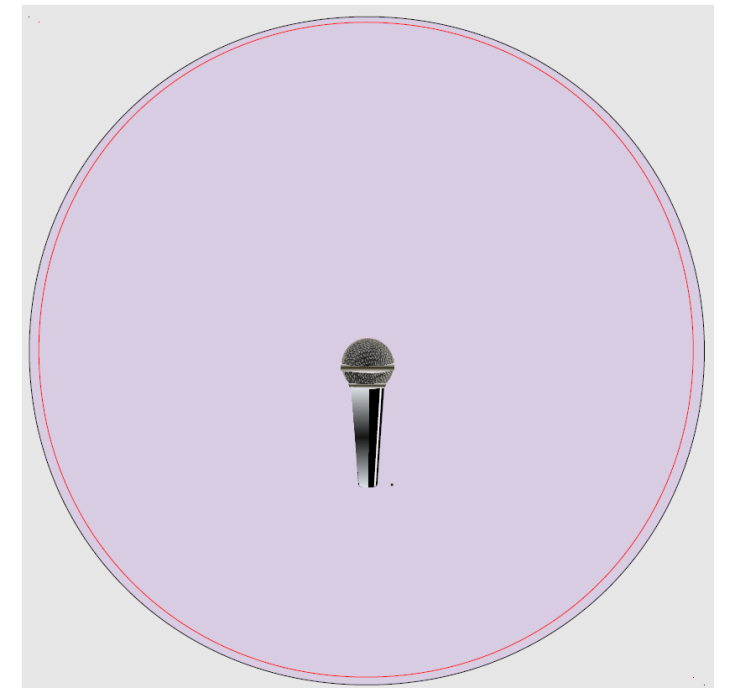
Sampling the Sphere

The systematic approach:

Find a set of microphones that cover the sphere completely.

Avoid redundant information.

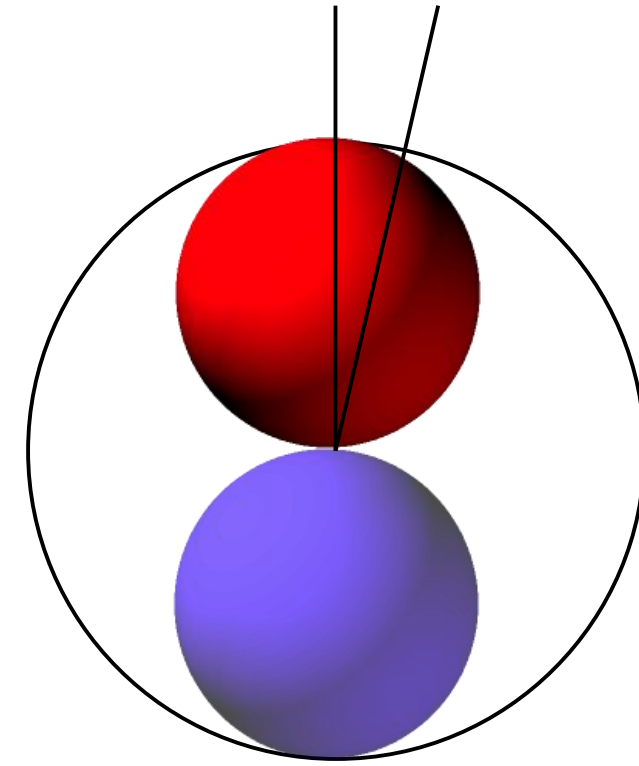
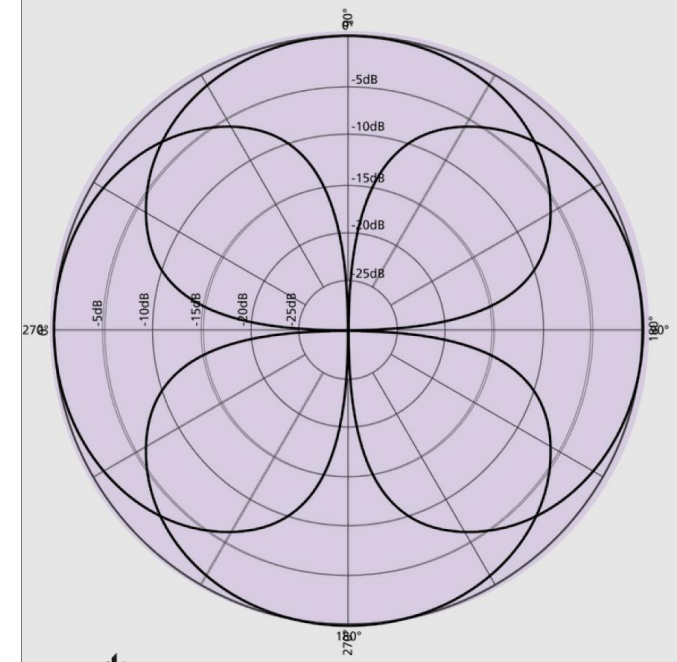
- Omnidirectional microphone = 0th order component
- **W** is the pressure component.
Remains when Ambisonics is played back in mono.



by Jörn Nettingsmeier, <http://stackingdwarves.net>

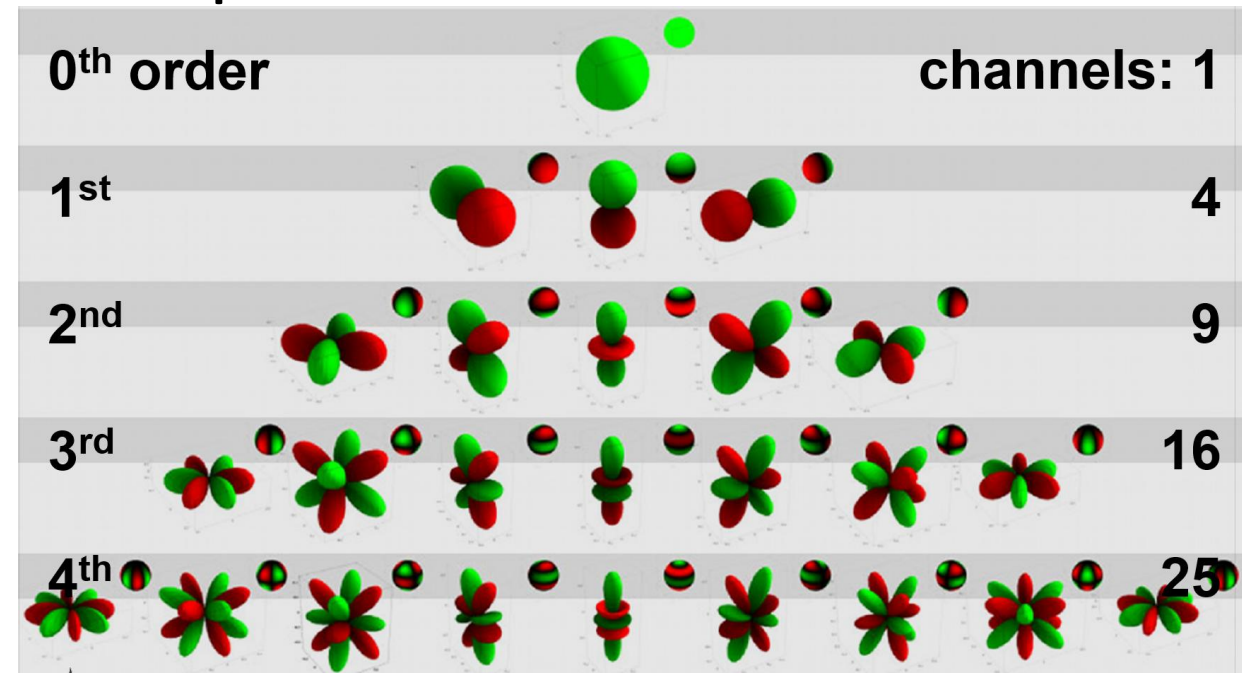
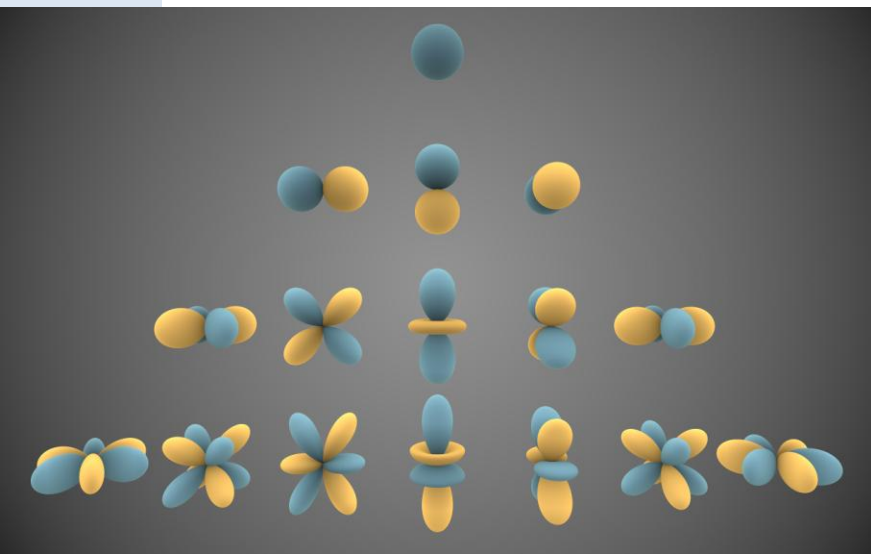
1st order Ambisonics

- Three figures-of-eight will also cover the sphere uniformly.
- Together with W, they provide directional information.
- In the B-format, they are denoted as X,Y,Z
- What's wrong with 1st order Ambisonics?
 - Low angular resolution:
off-axis sources have a shallow roll-off.
 - In practice, this means a small sweet spot
(= listening area)
- So we need narrower polar patterns which also sample the sphere uniformly and linearly independent



Spherical Harmonics: Higher-order Ambisonics

- **Spherical harmonics** are a set of functions used to represent functions on the surface of the **sphere**.
- They are a higher-dimensional analogy of Fourier series, which form a complete basis for the set of periodic functions of a single variable.



Ambisonics Defined

“**Ambisonics** goes further than stereo in that what it does... is to sample the acoustic field in such a way that the combination of the signals from all the loudspeakers in the array produces... a reconstruction of the original acoustic wave field (both traveling and standing wave components).

“If a listener puts their head in the sound field, because the wavefronts are similar to the original, the perception of directionality and space should correspond to the original too. It is a ‘wavefront reconstruction’ scheme in the small. The ear signals (crosstalk and all) will be correct if the reconstructed wavefronts are correct. This is just like natural hearing.

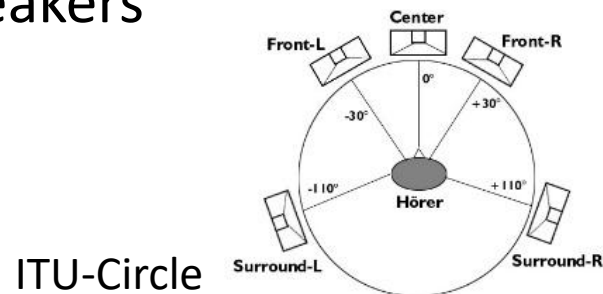
“Increasing the number of loudspeakers in Ambisonics (each fed its correctly-decoded signal) increases the accuracy of the reconstruction and the region over which it holds up. The interaural arrival-time differences also correspond to natural hearing in Ambisonics.”

—*Stanley Lipshitz*

5.1 and Ambisonics - Compared

5.1 surround

- Localization achieved solely with level
- Poor inter-speaker imaging except at front
- Localization varies with listener position
- Small sweet spot
- Horizontal surround uses six channels
- Speakers in special positions (e.g. ITU)
- One-to-one channel relationship from studio to speakers



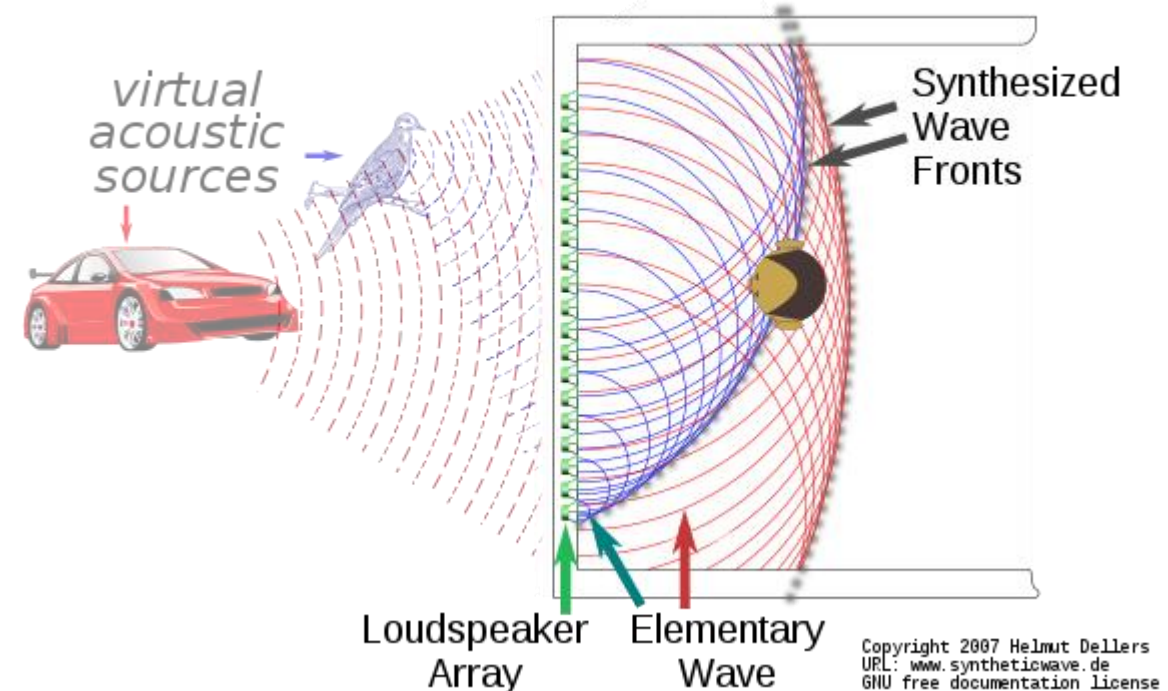
Ambisonics

- Localization includes other mechanisms
- Good inter-speaker imaging all round
- Less variation of image with position
- Larger sweet spot
- Even with height, only four channels needed (1st order)
- Decode to 5.1 or other configurations
- Recover B-Format and decode for completely flexible speaker arrays

Spatial Audio: Wave Field Synthesis (WFS)

- A spatial audio rendering technique, characterized by creation of virtual acoustic environments
- Produces artificial wavefronts synthesized by a large number of individually driven loudspeakers.
- The localization of virtual sources in WFS does not depend on or change with the listener's position

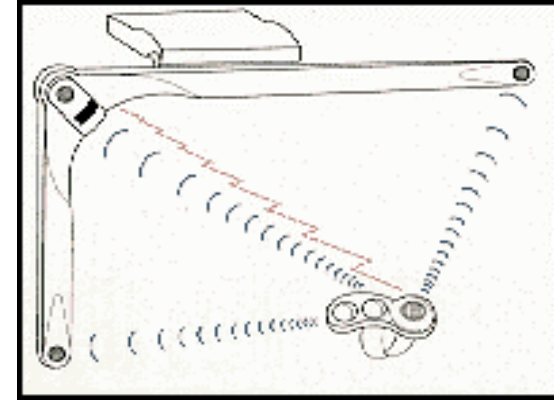
Wave Field Synthesis principle



MPEG-H 3D Audio (2013 -)

- Supports coding audio as **audio channels**, **audio objects**, or **higher order ambisonics** (HOA)
- Can support up to **64 loudspeaker channels** and **128 codec core channels**
- Audio objects may be used alone or in combination with channels or HOA components
- Use of audio objects allows **interactivity** or personalization of a program by adjusting gain or position of the objects during rendering in the decoder
- Audio is encoded using an improved modified discrete cosine transform (MDCT) algorithm
- Transmit immersive sound as well as mono, stereo, or surround sound.
- MPEG-H 3D Audio decoder renders bitstream to a number of standard speaker configurations as well as to misplaced speakers.
- **Binaural rendering** of sound for headphone listening is also supported
- Used since May 2017 in terrestrial South Korean UHD TV broadcast

Audio Processing Examples



- reshape impulse response to simulate a different room
- change perceived location from which sound comes
- locate speaker in 3D space using microphone arrays
- acoustic tracking
- cover missing samples
- echo cancellation
- Spatial audio (multi-speaker setups)
 - Audio objects
 - High number of channels



In Retrospect: Importance of Space “Delay Maschine” Gasometer

